សាកលវិទ្យាល័យភូមិន្ទភ្នំពេញ

ROYAL UNIVERSITY OF PHNOM PENH

ប្រធានបទនិក្ខេបបទ
ប្រព័ន្ធកម្មង់ម្ហូបតាម QR Code

A Research Report
In Partial Fulfilment of the Requirement for the Degree of Bachelor
Program of Information Technology Engineering

Im Phannyvath
Chea Kimheng

June 2025

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QR Code-Based Digital Menu System

A Research Report
In Partial Fulfilment of the Requirement for the Degree of Bachelor
Program of Information Technology Engineering

Im Phannyvath
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Examination committee: Mr.Chanpiseth Chap

March 2025

មូលន័យសង្ខេប

នៅក្នុងបរិបទនៃសាកលវិទ្យាល័យ ការបញ្ជាទិញអាហាររបស់និស្សិត និងបុគ្គលិកនៅតែពឹងផ្អែកលើវិធីសាស្ត្របែបប្រពៃណី ដូចជា ម៉ឺនុយក្រដាស និងការបញ្ជាពាក្យមាត់ ដែលបង្កើតឱ្យមានបញ្ហាដូចជា ការចំណាយពេលវេលា ការយល់ច្រឡំទៅការបញ្ជាទិញ និងការបាត់បង់ប្រសិទ្ធភាព ដែលបង្កឱ្យមានភាពមិនងាយស្រួល។ ដោយមិនមានប្រព័ន្ធឌីជីថលដែលអាចឆ្លើយតបនឹងតម្រូវការដោយផ្ទាល់បាន គម្រោងនេះត្រូវបានរៀបចំឡើង ដើម្បីបង្កើតប្រព័ន្ធម៉ឺនុយ QR កូដ ឌីជីថល ដែលអាចជួយសម្រួលដល់ដំណើរការបញ្ជា និងដឹកជញ្ជូនអាហារបានយ៉ាងមានប្រសិទ្ធភាព។ដូច្នេះជាការឆ្លើយតបទៅនឹងបញ្ហានេះ ការសិក្សានេះមានគោលបំណង បង្កើតQR កូដដើម្បីប្រើប្រាស់នៅក្នុងការកុម្មង់ម្ហូបអាហារនៅតាមហាងនានាជាការសម្រួលទៅដល់អតិថិជនទាំងពេលវេលា និងបទពិសោធន៍។ នាយុគសម័យឌីជីថលបច្ចុប្បន្ន ការប្រើប្រាស់បច្ចេកវិទ្យាបានចូលរួមចំណែកក្នុងការផ្លាស់ប្តូររបៀបនៃការរស់នៅរបស់មនុស្សជាច្រើន ទៅលើវិស័យផ្សេងៗជាពិសេសការរស់នៅប្រចាំថ្ងៃដូចជាការប្រើប្រាស់ក្នុងអាជីកម្មជាដើម។ ប្រព័ន្ធនេះបានបង្កើតឡើងដោយប្រើប្រាស់បច្ចេកវិទ្យា Web ទំនើប ដើម្បីផ្តល់បទពិសោធន៍ប្រើប្រាស់ដ៏ងាយស្រួល និងមានមុខងារជាច្រើនដូចជា ការបង្ហាញមីនុយជាក់ស្តែងដោយមានជម្រើសភាសាច្រើនសម្រាប់អ្នកប្រើប្រាស់, និងការបង្កើត QR code សម្រាប់ដាក់ទៅតាមតុនីមួយៗនៅក្នុងហាង។ ការធ្វើបែបនេះ អាចជួយឲ្យអ្នកលក់ម្ហូបធ្វើការបានលឿន និងមានប្រសិទ្ធភាព ខណៈពេលដែលអ្នកប្រើប្រាស់ទទួលបានបទពិសោធន៍ដ៏រលូន និងងាយស្រួល។ ក្រៅពីនេះ ប្រព័ន្ធនេះគ្រោងនឹងមានការបន្ថែមមុខងារនាពេលអនាគត ដូចជា ការទូទាត់តាមអនឡាញយ៉ាងមានសុវត្ថិភាព, ប្រព័ន្ធចែកចាយ និងការគ្រប់គ្រងអ្នកផ្គត់ផ្គង់សម្រាប់ម្ចាស់អាជីវកម្មនានានៅក្នុងស្រុក។ យោងទៅតាមការវាយតម្លៃទៅលើការប្រើប្រាស់ QR-Code សម្រាប់កុម្មង់អាហារ មតិយោបល់អ្នកប្រើ បង្ហាញថា មានអ្នកប្រើជាង 90% បានចាត់ទុកថាការប្រើប្រព័ន្ធនេះមានភាពងាយស្រួល

ក្នុងការប្រើប្រាស់។ ដូច្នេះការសិក្សានេះបានបង្ហាញពីសក្តានុពលក្នុងការដោះស្រាយ ក៏ដូចជាសម្រួលទៅដល់អតិថិជននិងម្ចាស់អាជីវម្មនានា ដើម្បីឱ្យអាជីវកម្មដំណើរការបានទៅរលូនប្រកបដោយប្រសិទ្ធភាព។

ABSTRACT

Within the university context, the food ordering process for students and staff still relies on traditional methods such as paper menus and verbal communication. These methods lead to several challenges, including time consumption, miscommunication in orders, reduced efficiency, and overall inconvenience. Without a digital system that directly addresses these issues, this project was initiated to develop a digital QR code menu system aimed at streamlining food ordering and delivery in a more efficient manner. To address this problem, the study aims to implement QR codes for food ordering at various vendor locations, with the goal of improving both time efficiency and the overall customer experience. In the current digital era, the use of technology plays a critical role in transforming daily life—especially in areas such as business and service delivery. The system was developed using modern web technologies to offer a user-friendly experience and includes several key features: real-time menu display, multilingual options for users, and dynamic QR code generation for each table within vendor areas. This approach allows vendors to operate more efficiently while providing a smooth and convenient experience for customers. Furthermore, the system is planned to expand in the future to include features such as secure online payment, delivery coordination, and supplier management tools for local business operators. According to evaluations of the QR code food ordering system, over 90% of users reported that it was easy to use. Therefore, this study highlights the potential of such a system to address existing challenges while enhancing the experience for both customers and business owners, enabling smoother and more effective operations.

SUPERVISOR'S RESEARCH SUPERVISION STATEMENT

TO WHOM IT MAY CONCERN

Name of program: Bachelor Degree of Engineering of Information Technology
Name of candidate: Im Phannyvath , Chea Kimheng

Title of research report: RUPP Qr-Menu

This is to certify that the research carried out for the above titled bachelor's research report was completed by the above-named candidate under my direct supervision. This thesis material has not been used for any other degree. We played the following part in the preparation of this research report: RUPP Qr-Menu

Supervisor's name: Chanpiseth Chap

Supervisor's signature: ……

Date: ……..……

CANDIDATE'S STATEMENT

TO WHOM IT MAY CONCERN

This is to certify that the research report that We, Im Phannyvath and Chea Kimheng Hereby present entitled for the degree of Bachelor of Information Technology Engineering at the Royal University of Phnom Penh is entirely my own work and, furthermore, that it has not been used to fulfill the requirements of any other qualification in whole or in part, at this or any other University or equivalent institution.

No reference to, or quotation from, this document may be made without the written approval of the author.

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Signed by Supervisor: Chap Chanpiseth

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Sincerely,
Im Phannyvath & Chea Kimheng

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CHAPTER 1 :INTRODUCTION

1.1 Background of the Study

The rapid advancement of technology has transformed the way people interact with services, including food ordering systems. In educational environments, such as universities, traditional methods of food ordering often involve physical menus, manual processes, and long waiting times, which can be inefficient and time-consuming. Recognizing the need for a more efficient solution, the Royal University of Phnom Penh (RUPP) QR-Menu project was developed to revolutionize the food ordering experience within the university.

The RUPP QR-Menu system enables students and staff to conveniently order food by simply scanning a QR code or accessing a link, eliminating the need for physical menus and providing a faster, more streamlined process. This system leverages modern technology to address common challenges faced in traditional food ordering, such as delays during peak hours and miscommunication between customers and vendors. By integrating QR codes and digital menus, the system promotes accessibility, accuracy, and operational efficiency, creating a user-friendly solution that benefits both customers and food vendors.

Historically, food ordering systems relied on direct communication, handwritten notes, or manual ticketing, which were prone to errors and inefficiencies. With the advent of digital technology, solutions such as mobile apps and online platforms have become increasingly popular in various industries. However, in academic settings, there remains a gap in leveraging such technology to address specific needs. The RUPP QR-Menu bridges this gap by providing a simple yet effective solution tailored to the university environment.

In addition to convenience, the QR-Menu system also aims to reduce paper waste and support environmental sustainability by replacing physical menus with digital alternatives. The system is designed to adapt to the growing demand for digital transformation in educational institutions, ensuring that students and staff have a seamless and efficient experience.

While the implementation of digital solutions like the QR-Menu presents opportunities, challenges such as system usability, scalability, and user adoption must also be addressed. By analyzing these factors and incorporating user feedback, the RUPP QR-Menu project seeks to serve as a model for leveraging technology to improve services in academic institutions. This study examines the development and impact of the QR-Menu system, emphasizing its potential to set new standards for efficiency, accessibility, and sustainability in food ordering processes.

1.2 Problem Statement

In Cambodia, online ordering systems are growing in popularity but often lack essential features and user-friendly interfaces, leading to inefficiencies and dissatisfaction. To address these challenges, I propose developing a **QR Code Menu System** for the **Royal University of Phnom Penh (RUPP)**. This system will streamline the ordering process by allowing students, staff, and visitors to scan QR codes for quick access to menus, place orders, and make payments.
Key features will include:

- **Unique QR codes** for each table or location on campus
- A **responsive and intuitive interface** with detailed menu descriptions, pricing, and images
- **Real-time updates** to reflect menu changes and availability
- **Order tracking** to monitor order status from preparation to delivery

Additionally, the system will introduce a **job application feature** that allows RUPP students to apply for roles such as delivery personnel or order management assistants. This not only offers employment opportunities but also fosters campus engagement and skill development.

Moreover, **suppliers—both internal and external—can apply to supply goods to the cafeteria**. For instance, if the **Agricultural Department** of RUPP wants to contribute by supplying fresh vegetables or other food products, they can register as official suppliers through the system. Their applications will be reviewed by the cafeteria team, and once approved, they can begin supplying goods regularly. This promotes a sustainable, community-driven ecosystem where local resources and student talent are effectively utilized.

Ultimately, this system aims to enhance convenience, improve operational efficiency, and deliver a modern, user-centered dining and service experience tailored to the needs of the RUPP community.

1.3 Objective of the Study

Of course, we decided to build this system due to the following important factors below:

- **To save time for students and staff**: The system simplifies the ordering process, making it quicker and more convenient for users at RUPP.
- **To create part-time job opportunities**: Students can work as delivery personnel, earning extra income while contributing to faster service.
- **To support cafeteria operations**: The system streamlines ordering and delivery, making it easier and more efficient for RUPP cafeterias to manage their services.
- **To enhance user experience**: By providing a seamless and user-friendly platform, the system improves convenience and satisfaction for everyone involved.

1.4 Limitation and Scope

This research focuses on developing a system to save time for students and staff at the Royal University of Phnom Penh (RUPP) by implementing a **QR Code Menu system**. The system enables users to scan QR codes to access menus, place orders, and make payments efficiently. Additionally, the project provides a feature where students can take part-time jobs as delivery personnel, contributing to faster service and supporting cafeteria operations. While the project has clear goals and potential, there are several limitations and defined scopes to consider.

The scope of the project is limited to the RUPP campus, focusing exclusively on the university's cafeterias. The system is designed primarily for ordering food and beverages, and its functionality may not extend to more complex services such as inventory management or advanced analytics for vendors. Moreover, the system's success depends on user adoption, meaning students, staff, and cafeteria vendors must actively use the platform for it to be effective.

The project also has certain limitations. For instance, it may face challenges with internet connectivity, as the system requires stable access for QR code scanning and order processing. Additionally, the delivery feature is limited to part-time student workers, which could lead to delays during peak hours or when there is a shortage of available delivery personnel. Another

limitation is that the system may initially struggle to gain the trust of users who are unfamiliar with digital ordering systems, requiring proper onboarding and awareness campaigns.

Furthermore, the system is designed for a single-class environment (RUPP) and may not be easily scalable to other institutions without customization. These limitations highlight the need for continuous refinement and future enhancements to improve its reliability, scalability, and overall performance.

1.5 Structure of Study

This study is structured into five chapters, organized as follows:

- **Chapter 1**: Introduces the background knowledge on the QR Code Menu system for RUPP, highlighting its purpose, objectives, and the challenges it addresses.
- **Chapter 2**: Provides a review of related systems and studies, focusing on the integration of QR codes and digital ordering systems, as well as their impact on time-saving and user convenience.
- **Chapter 3**: Describes the methodology of the study, including system design, feature development, and the approach used to create the part-time delivery feature for students.
- **Chapter 4**: Discusses the implementation of the system, detailing the technologies and tools used, and presents the results from testing the system's functionality in real-world scenarios.
- **Chapter 5**: Contains concluding remarks, summarizes the findings of the study, and outlines recommendations for future improvements and scalability of the system

.

Chapter 2 : LITURATURE REVIEW

2.1 History of Food Ordering Systems

The evolution of food ordering systems reflects a broader transition from manual, labor-intensive processes to digitized, user-centered platforms. Traditionally, food ordering relied on verbal communication or handwritten notepads, which were susceptible to delays, inaccuracies, and miscommunication—especially during peak service hours. These early methods were functional but not scalable and posed challenges in high-traffic environments such as cafeterias, where speed and clarity were essential.

The introduction of Point-of-Sale (POS) systems in the 1990s marked the beginning of digital transformation in the food industry. POS terminals allowed restaurant staff to input customer orders electronically, improving speed and enabling real-time transmission to kitchen staff. However, these systems remained highly dependent on hardware infrastructure and required significant investment, making them less suitable for low-budget institutions such as schools or public canteens.

In the early 2010s, the proliferation of mobile devices and wireless internet access gave rise to mobile food ordering applications. These platforms enabled customers to browse menus, place customized orders, and make digital payments through user-friendly interfaces. Global services like Foodpanda and Uber Eats revolutionized the consumer experience but also introduced service fees, data privacy concerns, and system complexity that limited their use in closed ecosystems like educational institutions.

The emergence of QR code-based digital menus in the mid-2010s offered a cost-effective, scalable, and contactless alternative to both traditional and app-based systems. Initially used as a novelty or supplementary tool, QR menus became mainstream during the COVID-19 pandemic when restaurants, cafés, and universities sought safer ways to serve customers. Customers could scan a printed QR code using their smartphone camera to instantly access the menu, place orders, and complete payment—without needing to download any apps or touch shared surfaces.

Importantly, QR code ordering systems align well with the operational and financial constraints of public universities in developing countries. Studies have shown that QR-based systems improve operational efficiency while requiring minimal infrastructure. They also support user autonomy and reduce the need for trained ordering staff, making them ideal for student-driven

environments. Furthermore, they provide an opportunity for academic institutions to integrate digital skills training, sustainability goals, and on-campus job creation within a single digital platform.

Thus, the history of food ordering systems—from verbal to digital, and from centralized POS to decentralized QR access—sets the stage for solutions like the RUPP QR-Menu. Designed with the university's ecosystem in mind, this system embodies the next step in food service innovation by offering real-time updates, campus-based delivery, and supplier integration in a resource-conscious, inclusive, and student-centric framework.

2.2 Related and Previous Works

Over the past few years, there has been a growing trend of implementing QR code technology in various sectors, particularly in the foodservice industry. The use of QR codes has revolutionized the way customers interact with restaurants, allowing for a more seamless and efficient ordering experience. This section highlights the studies and innovations that have contributed to the development of QR code-based menu systems and their impact on customer service and operational efficiency.

Jain et al. (2021) explored the role of QR codes in modern restaurant services, emphasizing the convenience of using mobile devices for contactless ordering. Their research indicated that QR code menus are not only more hygienic but also contribute to reduced wait times, as customers can directly place orders from their smartphones. This feature became particularly important during the COVID-19 pandemic, where contactless interactions were prioritized. Their study also identified several challenges, such as ensuring all customers have access to smartphones and managing issues related to mobile device compatibility.

Lee and Chang (2020) studied the implementation of QR code-based ordering systems in small and medium-sized restaurants, focusing on improving the speed and accuracy of orders. They noted that QR code systems are particularly useful for restaurants with high customer turnover, as they eliminate the need for waitstaff to manually take orders. By allowing customers to order directly from their phones, restaurants can significantly reduce errors in the ordering

process, leading to increased customer satisfaction. Furthermore, the integration of payment options within QR code systems allows for a faster, more secure transaction process.

Kumar et al. (2022) examined the impact of QR codes on restaurant management efficiency, particularly in relation to order and inventory management. Their study found that by integrating QR code menus with a centralized order management system, restaurant staff can easily track inventory levels, reducing food waste and preventing stockouts. This system also streamlines the kitchen workflow, as orders are automatically sent to the kitchen upon submission, allowing staff to focus on food preparation instead of order taking.

Singh et al. (2023) focused on the integration of QR code menus with additional services, such as delivery management and real-time updates. Their research highlighted a QR code system that allowed restaurant owners to hire part-time delivery personnel, providing students or local workers with flexible employment opportunities while also ensuring efficient delivery services. This model not only increased delivery speed but also helped restaurants manage labor costs during peak hours. Singh's study demonstrated that such integrations help improve the overall customer experience and create more business opportunities for individuals seeking part-time work.

In addition, **Patel (2021)** introduced the concept of using QR code menus as part of a broader smart restaurant ecosystem, where data collected from customer orders can be analyzed to personalize the dining experience. By leveraging customer preferences and behavioral data, restaurant owners can tailor their menu offerings, introduce targeted promotions, and enhance customer engagement.

These studies illustrate the wide range of applications for QR code menus in the restaurant industry, highlighting their ability to improve operational efficiency, reduce wait times, and offer new employment opportunities. Building on these insights, our project aims to develop a QR code menu system tailored for RUPP, providing not only a user-friendly ordering system but also the option for students and staff to participate in the delivery process, thereby enhancing overall convenience and saving time

References	Approach	Data source	Findings
(Jain et al., 2021)	QR code-based ordering system	Restaurant case study	QR code menus offer convenience, hygiene, and reduced wait times. They are especially useful during the COVID-19 pandemic for contactless ordering.
(Lee & Chang, 2020)	QR code-based ordering and direct payment systems	Small and medium-sized restaurants	QR code menus improve speed and accuracy of orders, reduce errors, and streamline payment, especially in restaurants with high customer turnover.
(Kumar et al., 2022)	QR code integration with inventory and order management	Restaurant management system	QR code systems improve inventory tracking, reduce food waste, and streamline kitchen workflow by automatically sending orders to the kitchen.
(Singh et al., 2023)	QR code-based menu with delivery management feature	Restaurant case study	QR code menus enable flexible employment for delivery personnel enhancing overall customer experience.
(Patel, 2021)	QR code menus integrated with smart restaurant ecosystem	Smart restaurant setup	QR code menus allow for data analysis on customer behavior, enabling personalized menu offerings, promotions,

Table 1:Table 1: Summary of Studies on QR Code Menus and Their Applications in the Foodservice Industry

2.2.1 Critical Analysis of Related Works

The existing body of literature on QR code-based ordering systems offers valuable insights into technological adoption in the food service sector; however, most studies focus on commercial or restaurant settings, leaving a gap in research related to academic or campus-based environments. For instance, Jain et al. (2021) emphasized the hygienic advantages and reduced wait times of QR code systems during the COVID-19 pandemic. While this is relevant, their study assumes a customer base that is already familiar with smartphone-based ordering systems, which may not apply in educational institutions where digital literacy varies among users. Similarly, Lee and Chang (2020) analyzed QR code systems in small and medium-sized restaurants, noting efficiency in direct payments and order accuracy, but they did not consider use cases involving shared campus infrastructure or multilingual interfaces necessary for inclusive student engagement.

Kumar et al. (2022) focused on integrating QR code menus with inventory management systems to optimize kitchen workflows. Although their findings demonstrate operational benefits for professional kitchens, their model presumes a level of logistical and technical capacity not always available in university cafeterias. Furthermore, Singh et al. (2023) introduced a delivery system involving part-time staff, which shares similarities with the RUPP QR-Menu's student delivery feature. However, their research did not explore how such systems could support student employment or skill development within an academic context.

Additionally, Patel (2021) discussed smart restaurant ecosystems leveraging QR code data to offer personalized dining experiences and targeted marketing, but this commercial personalization model is difficult to adapt in campus settings with limited budgets and a diverse user base. These gaps highlight the need for a system that not only enables efficient food ordering but also supports local operations and student participation.

The RUPP QR-Menu system builds upon these studies by contextualizing QR code technology for a university setting. It introduces key innovations such as part-time job creation for students, localized supplier registration (e.g., internal departments like agriculture), and a multilingual interface to accommodate all users on campus. By integrating academic needs with operational efficiency, the system addresses shortcomings in existing research and provides a scalable, student-centric solution. Thus, this project contributes to the literature by demonstrating how QR-based systems can be effectively adapted to educational institutions, promoting inclusivity, sustainability, and digital transformation in low-resource environments.

2.3 State of the Art

2.3.1 QR Code Menus: Revolutionizing Dining and Ordering Systems

QR code menus have emerged as a game-changer in the hospitality industry, particularly in response to the increased demand for digital, contactless solutions. The rise of QR code technology has transformed how customers interact with menus, place orders, and pay bills. These menus are typically accessed by scanning a QR code placed on the table, on walls, or at service points within a restaurant or café. Upon scanning, customers are redirected to a digital menu on their smartphone, which offers a detailed, user-friendly interface where they can browse items, view descriptions, and place their orders.

The appeal of QR code menus is clear: they streamline the ordering process, reduce dependency on physical menus, and enhance operational efficiency. Restaurant owners or canteen operators can update menu offerings and prices instantly, eliminating the need to reprint physical menus. Additionally, the system supports real-time menu updates and order tracking, which improves both customer satisfaction and backend operations.

Beyond improving the dining experience, QR code menus create opportunities for community involvement and student engagement, particularly in institutional settings such as the Royal University of Phnom Penh (RUPP). For instance, the system enables **students to apply for on-campus delivery jobs**, helping them earn income while developing soft skills such as time management, communication, and customer service. Orders placed via the QR code menu are automatically sent to a central dashboard, where student delivery personnel are assigned tasks and can deliver meals to the appropriate table or area within the university.

Moreover, the system is designed to accommodate **suppliers**, including university departments such as the **Faculty of Agricultural Sciences**, which may wish to provide fresh vegetables, fruits, or other organic produce. These departments or local vendors can **apply to become official suppliers through the platform**, submit their product lists, and participate in the university's food supply chain. This not only supports sustainability and local sourcing but also gives students from agriculture and business faculties hands-on experience in supply chain management, marketing, and logistic.

1.Dashboard to menu flow

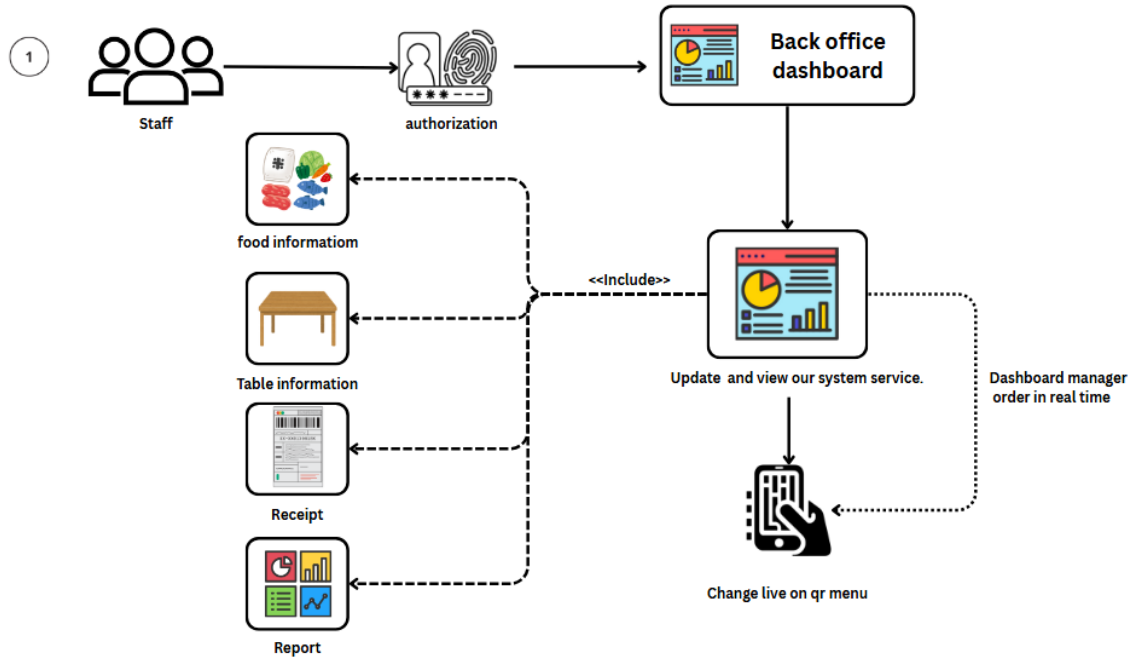


Figure 1: Staff Process flow

1. Staff Authorization

The process begins when a cafeteria staff member initiates access to the system. They are required to log in through an authorization step using credentials or biometric data. This ensures secure access and prevents unauthorized changes to system content or operations.

2. Access Back Office Dashboard

After successful login, the staff is directed to the **Back Office Dashboard**, which serves as the central control panel for managing the QR Menu System. This dashboard provides real-time visibility into all system operations, including order flow, menu content, and service configuration.

3. Perform Administrative Services (Food, Tables, Receipts, Reports)

Inside the dashboard, staff carry out all essential administrative functions from a unified interface. They can update food information such as item names, availability, and pricing; manage table configurations including QR code assignments; generate and review customer receipts; and access performance reports related to orders and service efficiency. All these tasks are handled

under a single service module, streamlining the process and reducing complexity for staff. This ensures consistency, quick updates, and accurate data flow throughout the system.

4. Manage Orders in Real Time

The dashboard also enables staff to monitor and manage customer orders in real time. As orders are submitted from the customer-facing menu website, they appear instantly on the dashboard. Staff can update each order's status — such as "Received", "Cooking", "Ready", or "Delivered" — and track its progress through the fulfillment lifecycle. This improves coordination between the kitchen and service staff.

5. Apply Live Changes to QR Menu

Any updates made by staff — whether related to food items, table assignments, or order details — are immediately reflected on the **QR Menu interface** that customers access by scanning codes at their tables. This real-time synchronization ensures that users always see accurate, up-to-date information, and that the digital menu responds dynamically to staff inputs.

2.App process flow

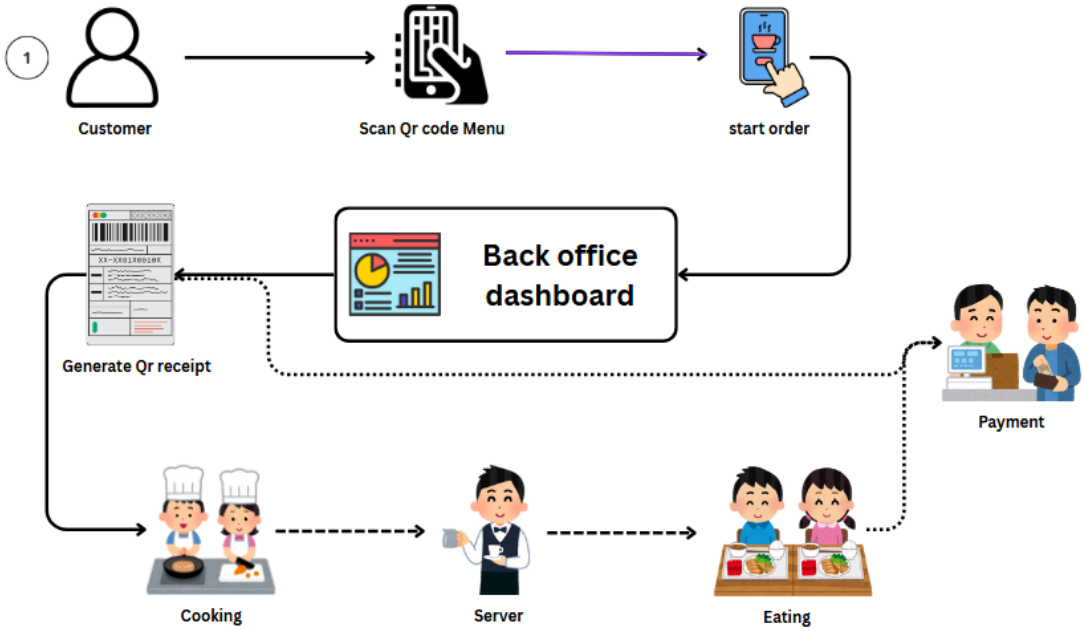


Figure 2: Customer Process Flow

1. Customer Interaction

The ordering journey begins when a customer, typically a student, staff member, or visitor, approaches a table equipped with a QR code. Using their smartphone, the customer scans the QR code placed on the table. This action redirects them to the digital menu interface, which is built

using Nuxt.js and Vue.js on the frontend. The QR code contains encoded information about the specific table ID, ensuring that any order placed will be correctly linked to that location. This provides a seamless and personalized entry point into the system.

2. Start Order

After accessing the digital menu, the customer is presented with a list of food items that includes names, prices, availability, and images. The interface allows the customer to browse at their own pace, select items, customize them (if applicable), and proceed to checkout. Once the order is finalized, the system sends a request containing the selected items and user input to the backend API. This backend is built with Express.js and processes the request by validating the data, generating a unique order ID, and saving the order information to a MongoDB Atlas database. A confirmation is then sent back to the frontend to inform the user that the order has been successfully placed.

3. Back Office Dashboard

As soon as the order is recorded in the system, it becomes visible on the back office dashboard used by cafeteria staff. This dashboard is a centralized management interface that displays all incoming orders in real time. Staff can view essential details such as the order contents, associated table number, order time, and any special instructions provided by the customer. The dashboard also allows staff to update the status of each order as it progresses through stages like "Received", "Cooking", "Ready", and "Delivered". This status tracking ensures that both staff and customers remain informed throughout the order lifecycle.

4. Generate QR Receipt

Once the order is accepted, the system generates a digital receipt containing key order information, including the order ID, table number, itemized list, prices, and total cost. This receipt may also include a QR code that links back to the order status for verification or recordkeeping purposes. Depending on the setup, the receipt can be printed for physical use or displayed digitally on the customer's device or the kitchen's screen. This receipt plays an important role in confirming the order and ensuring traceability.

5. Cooking

With the order confirmed, the kitchen staff begins preparation. They rely on the back office dashboard to track the list of incoming orders, and once a specific order is ready to be cooked, they update its status to "Cooking". This helps keep the process organized and ensures visibility for all

staff members involved. The food is then prepared according to the customer's specifications. When the cooking is complete, the status is updated once again—this time to “Ready”—indicating that the meal is available for pickup or serving.

6. Server (Delivery Staff / Waiter)

When the order is marked as “Ready”, it signals the server—often a student worker or designated staff member—to collect the food from the kitchen. The server receives notification of the task via the system and is responsible for delivering the correct order to the corresponding table, based on the order’s table ID. After successfully delivering the food, the server can mark the order as “Delivered” on the dashboard. This handoff closes the loop between kitchen and customer while ensuring order accuracy and accountability.

7. Eating

With the food delivered, the customer begins eating. This marks the final user-facing step in the order process. Depending on the payment model, the customer may proceed to complete the transaction either before eating (prepaid) or after eating (postpaid) via cash or digital payment systems. The system captures this status update, marking the order as “Completed” or “Paid”, which finalizes the process. The structured flow from ordering to eating provides a modern, efficient, and user-friendly dining experience that minimizes delays and maximizes satisfaction for both customers and cafeteria staff.

8. Payment

The final and crucial step in the ordering process is payment. Depending on the system configuration, payment can occur either immediately after placing the order (prepaid) or after the food has been delivered and consumed (postpaid). Once the customer completes the order and confirms the selection, they are directed to the payment interface. In future versions of the system, this may include integration with popular Cambodian payment platforms such as Wing, Pi Pay, or Bakong. Upon successful transaction, the backend updates the payment status of the order to “Paid” in the database, and a confirmation is reflected on both the customer’s interface and the back office dashboard. This ensures accountability and transparency for staff, while giving the customer confidence that their order has been fully processed. If payment is done manually (e.g., cash at the counter), staff can manually update the order status in the dashboard after receiving payment.

2.3.2 Customer Experience: The Role of QR Code Menus

QR code menus are not just a technological advancement; they significantly enhance the overall dining experience for customers. The primary benefit of these digital menus is their convenience, offering customers a faster and more efficient way to place orders. With the ability to browse the menu at their own pace, customers feel less rushed, leading to higher satisfaction. Additionally, QR code menus contribute to a contactless dining experience, which became especially important during the COVID-19 pandemic. In addition to reducing physical contact, QR code menus provide several other advantages:

- **Faster Service**: Customers can quickly access the menu, place orders, and pay without waiting for a server to deliver a physical menu or process payments.
- **Real-Time Updates**: Restaurant owners can update the menu offerings or prices in real time, ensuring the menu is always current.
- **Customization**: Digital menus allow for greater customization options for customers, from dietary preferences to meal modifications, ensuring a more personalized dining experience.
- **Increased Comfort**: The ability to browse the menu privately and at one's own pace creates a more comfortable and relaxed atmosphere, making customers feel more at ease.
- **Reduction in Paper Usage**: By moving to digital menus, restaurants reduce their reliance on paper menus, contributing to environmental sustainability and cutting down on paper waste.
- **Energy Savings**: With fewer physical menus to print, clean, and maintain, restaurants can reduce operational costs and energy consumption.
- **More Time to Decide**: Customers can take their time to review the menu and make decisions without feeling rushed, which can lead to better decision-making and greater satisfaction with their meal choices.

Benefit	Impact on Customer	Customer Feedback
Reduced Wait Time	Customers place orders directly, speeding up the process.	Faster service leads to happier customers.
Contactless Interaction	Reduces physical contact between customers and staff	Increased satisfaction, especially post-pandemic.
Real-Time Updates	Menus can be updated quickly and easily, reflecting changes in prices or items.	Accurate, up-to-date information for customers.
Personalized Experience	Customers can customize their orders directly, catering to dietary preferences.	More personalized and flexible meal options.
Increased Comfort	Customers can browse the menu at their own pace, avoiding a rushed dining experience.	Customers feel more relaxed and less rushed during dining.
Reduced Paper Usage	Less reliance on physical menus, helping to reduce waste and promote sustainability.	Customers appreciate the eco-friendly approach.
Energy and Cost Savings	Less need for printing and managing physical menus, saving on operational costs.	Customers value the restaurant's resource efficiency.

Table 2:Key Benefits of QR Code Menus

2.3.3 Applier Experience: The Role of Student Candidates and Suppliers

The QR Code Menu system not only modernizes food ordering but also opens new opportunities for student employment and departmental collaboration within the university. Two key roles that support this system are **Delivery Job Appliers** and **Suppliers**.

Student Candidates as Delivery Staff

The system provides on-campus job opportunities for students to work as delivery personnel. After customers place their orders through the QR code menu, student workers receive the orders via a mobile-friendly dashboard and handle the delivery directly to specific tables or locations.

Key Benefits for Delivery Job Appliers:

- **Flexible On-Campus Jobs:** Students can apply for part-time delivery roles that accommodate their class schedules.
- **Skill Development:** These roles build communication, time management, and responsibility—skills valuable in future employment.
- **Convenient Application Process:** Students can apply for delivery jobs directly through the platform, track their application status, and get updates instantly.
- **Earn While Learning:** Paid opportunities help students financially without needing to leave campus.

Suppliers: Enabling Departments to Contribute

The platform also supports suppliers—particularly from departments like **Agriculture**, which may wish to provide fresh produce such as vegetables, fruits, or organic items to the cafeteria. Interested departments or individuals can **apply to be suppliers** through the system, and upon approval, they can list available items for procurement.

Key Benefits for Suppliers (e.g., Agriculture Department):

- **Promotes Sustainability:** Encourages local, organic, and fresh food sourcing directly from campus activities.
- **Hands-On Educational Experience:** Agriculture students gain real experience in food supply chains, product marketing, and logistics.
- **Income Generation for Departments:** By becoming suppliers, departments can support operational costs and fund student projects.
- **Easy Management:** Suppliers can manage item listings, availability, and delivery schedules through an integrated dashboard.

Role	Description	

		Benefit to RUPP Community
Student staff	Customers place orders directly, speeding up the process.	Faster service leads to happier customers.
Supplier	Reduces physical contact between customers and staff	Increased satisfaction, especially post-pandemic.

Table 3:Student staff and supplier

2.3.4 Integration with Technology: Enhancing Efficiency

In addition to enhancing customer experience, QR code menus also provide businesses with opportunities to streamline operations. Many QR code systems are integrated with point-of-sale (POS) systems, allowing for a seamless flow from the customer order to the kitchen. This reduces the chances of errors and ensures that orders are processed efficiently.

Moreover, the integration of data analytics in QR code menu systems enables businesses to gather valuable insights, such as customer preferences, ordering trends, and peak times. This data helps improve decision-making and enables restaurants to optimize their offerings.

CHAPTER 3 : FUNCTIONAL AND NON-FUNCTIONAL

The **Functional and Non-Functional Requirements** chapter serves as a foundational component of your research paper, defining the scope and expectations for the RUPP QR Code Menu system. It outlines:

- **What the system does** (functional requirements): The specific features and capabilities, such as scanning QR codes, placing orders, or managing deliveries.
- **How well the system performs** (non-functional requirements): The quality attributes, such as speed, security, usability, and scalability, that ensure the system is reliable and user-friendly.

This chapter defines the functional and non-functional requirements for the RUPP QR Code Menu system, providing a clear specification of its features and performance standards. Functional requirements outline the system's core functionalities, while non-functional requirements specify quality attributes such as performance, security, and usability. These requirements ensure the system meets the needs of students, staff, cafeteria vendors, delivery personnel, and suppliers within the Royal University of Phnom Penh's (RUPP) cafeteria environment, aligning with the project's objectives of streamlining food ordering, creating job opportunities, and enhancing operational efficiency.

3.1 Functional Requirements

Functional requirements describe the specific features and capabilities the system must provide to support efficient food ordering and management at RUPP. These are categorized by user interactions and system operations.

1. Table Creation and QR Code Generation:

The system shall allow cafeteria staff to dynamically create tables via the dashboard.

Each table shall automatically be assigned a unique QR code upon creation.

Scanning the QR code shall link the customer to the digital menu associated with that table.

Purpose: Streamlines table-level management, supports seamless ordering per location, and enables future features like table-based online payments.

2. QR Code Menu Access:

The system shall display a digital menu with food names, images, prices, and availability when a QR code is scanned.

The menu shall support real-time updates from the backend.

Purpose: Enables contactless and efficient access to menus, reduces paper waste, and improves accessibility during peak hours.

3. Multilingual Interface:

The system shall support switching between multiple languages, including Khmer and English.

Purpose: Enhances usability for local users and ensures inclusivity across the campus community.

4. Order Placement:

Users shall be able to select and customize menu items and add them to a cart.

The system shall support eat-in, pickup, and delivery options.

Purpose: Improves user convenience and adapts to diverse dining preferences within the university.

5. Payment Handling (Planned Feature):

The system shall be designed to integrate with local payment providers (e.g., Wing, Pi Pay, Bakong) in future iterations.

Upon integration, users shall receive confirmation after successful transactions. Purpose: Lays the foundation for secure and convenient cashless transactions, enhancing the digital ordering experience.

6. Vendor Menu & Order Management:

Vendors shall manage menu items (add, edit, remove) through a dashboard.

The system shall display real-time order status updates (e.g., Received, Preparing, Ready).

Purpose: Increases operational efficiency, minimizes manual errors, and ensures accurate service delivery.

7. Delivery Task Management:

Registered student personnel shall receive delivery assignments through the dashboard.

Delivery users can accept tasks, access delivery locations (e.g., room number), and update order status as delivered.

Purpose: Provides flexible, on-campus part-time job opportunities and improves delivery service during high-demand periods.

8. Supplier Integration:

Suppliers (e.g., Agricultural Department or external vendors) can apply to provide food items.

Approved suppliers can manage listings, receive requests, and track fulfillment records.

Purpose: Promotes local and campus-based sourcing, supports sustainability, and offers students hands-on supply chain experience.

3.2 Non-Functional Requirements

Non-functional requirements specify the system's quality attributes, ensuring it is reliable, user-friendly, and suitable for RUPP's cafeteria environment.

1. **Performance**:
    - The system shall support up to 300 concurrent users with a response time of under 2 seconds for menu loading and order submission.
    - Real-time order updates shall be delivered within 1 second.
    - **Purpose**: Ensures responsiveness during peak cafeteria hours (e.g., lunch rush rush), addressing delays noted in Chapter 1.1. The 300-user target reflects RUPP's typical cafeteria traffic.
2. **Scalability**:
    - The system shall accommodate additional cafeterias or campuses with minimal reconfiguration.
    - The database shall support up to 3,000 menu items and 300 daily orders.

- **Purpose**: Supports future expansion (Chapter 7) and ensures the system can handle growth without major redesign.

3. **Security**:
    - User data (e.g., payment details, passwords) shall be encrypted using AES-256 and JSON Web Tokens (JWT).
    - The system shall comply with PCI DSS standards for payment processing.
    - **Purpose**: Protects sensitive data, critical for user trust, especially with future payment integration (Chapter 7).
4. **Usability**:
    - The interface shall be mobile-optimized for devices with screen sizes from 320px to 1920px.
    - The system shall achieve a System Usability Scale (SUS) score of at least 80.
    - **Purpose**: Ensures accessibility for all users, aligning with the 90% usability satisfaction reported (Abstract). Mobile optimization supports RUPP's smartphone-reliant user base.
5. **Reliability**:
    - The system shall maintain 99.9% uptime, with daily automated backups via MongoDB Atlas.
    - Recovery from failures shall occur within 5 minutes.
    - **Purpose**: Minimizes disruptions, crucial for cafeteria operations during peak times, addressing connectivity limitations (Chapter 1.4).
6. **Accessibility**:
    - The interface shall meet WCAG 2.1 Level AA standards for users with disabilities.
    - **Purpose**: Promotes inclusivity, ensuring the system is usable by all RUPP community members, including those with visual or motor impairments.
7. **Maintainability**:
    - The system shall use modular code, enabling updates within 1 week by a single developer.
    - **Purpose**: Facilitates ongoing maintenance and future enhancements (Chapter 7), ensuring long-term sustainability.

CHAPTER 4 : MATERIALS

The materials section comprehensively describes the tools, technologies, frameworks, and resources utilized to design and implement the QR Code Menu project. This endeavor was built using the MERN stack for backend development and Nuxt.js for the frontend. Each component of the stack and associated resources played a vital role in ensuring the efficiency, scalability, and user-friendliness of the application.

4.1 Development Tools and Technologies

4.1.1 Frontend

- **Nuxt.js**: Nuxt.js was the cornerstone of the frontend development, selected for its robust server-side rendering (SSR) capabilities, seamless integration with Vue.js, and built-in routing and state management. It ensured a fast-loading, SEO-optimized, and dynamic user experience. Nuxt's modular architecture allowed for the inclusion of plugins and modules to enhance functionality.
- **Tailwind CSS**: The utility-first CSS framework, Tailwind CSS, provided a highly customizable and efficient approach to styling. Its extensive class-based design system facilitated the rapid development of a responsive and visually appealing interface while adhering to modern web design principles.
- **ShadCN UI**: ShadCN UI brought pre-styled, accessible, and customizable components to the project. This ensured a consistent design language and accelerated the development of complex UI elements such as modals, dropdowns, and form components.
- **Vue.js**: As the core JavaScript framework underlying Nuxt.js, Vue.js played a crucial role in building reactive and reusable UI components. Its simplicity, flexibility, and two-way data binding model made it ideal for developing interactive user interfaces. Vue's ecosystem, including Vuex for state management and Vue Router for navigation, contributed to a streamlined and efficient development process.

Figure 3: Frontend Technology

4.1.2 Backend

- **MongoDB**: The database layer leveraged MongoDB's NoSQL architecture, which was ideal for handling the semi-structured data associated with menu items, user profiles, and QR code metadata. The use of MongoDB Atlas offered automated scaling, backups, and real-time performance monitoring.
- **Express.js**: Express.js acted as the web application framework, enabling the creation of a lightweight and flexible API layer. Its middleware ecosystem provided robust support for request parsing, authentication, and error handling.
- **Node.js**: Node.js powered the server-side runtime environment, facilitating asynchronous, non-blocking operations that improved the backend's performance and scalability. Its compatibility with JavaScript ensured seamless communication between the frontend and backend.
- **Authentication Tools**: JSON Web Tokens (JWT) were employed to establish a secure authentication mechanism. JWT provided token-based stateless authentication, ensuring that user sessions remained secure while reducing server-side overhead.
- **Postman**: Postman was an indispensable tool for API testing, allowing for comprehensive verification of endpoint functionality, performance, and reliability. It provided features

such as automated testing, request chaining, and environment variable management to streamline the testing process.

- **Swagger**: Swagger was utilized for API documentation and testing, offering an interactive UI to explore and verify API endpoints. It enhanced developer collaboration, improved API visibility, and ensured consistency across the documentation process.

Figure 4: Backend Technology

CHAPTER 5 : METHODS

5.1 Research and Planning

5.1.1 Research

- **Target Audience**: The primary users of the cafeteria ordering system are students and teachers. It is important to understand their needs, such as quick and easy ordering, payment flexibility, and real-time updates.
- **Technology Solutions**:
  - **QR Code Integration**: QR codes will direct users to the cafeteria's menu, providing a seamless entry point.
  - **Payment Integration**: Research on integrating popular payment methods (QR code-based, online payments, etc.) that ensure ease of use and security.
  - **Real-time Order Tracking**: Explore solutions for tracking order status (e.g., WebSockets or Firebase) to update users in real time.
- **Competitor Analysis**: Review existing cafeteria ordering systems to identify strengths and weaknesses, ensuring your system is user-friendly and efficient.
- **User Feedback**: Conduct surveys with students and teachers to gather insights on their preferences for menu selection, payment options, and order tracking.

5.1.2 Planning

User Flow: The flow starts with users scanning the QR code to access the menu, selecting food items, confirming order preferences (Eat-in, Pick-up, Delivery), tracking the order, completing payment, and optionally leaving feedback.

Key Features:

- **Real-time Order Updates**: The system will display order status (Received, Cooking, Cooked, Delivered) in real time using ensuring users stay informed without refreshing the page.
- **Delivery Management**: A system will assign student workers for deliveries along with monitoring earnings for part-time roles.

- **Feedback Mechanism**: Users can leave ratings and comments on food, delivery, and service, helping improve the cafeteria's offerings based on customer input.

System Design:

- **Frontend**: A mobile-optimized, user-friendly interface that allows easy navigation, order customization, and real-time tracking.
- **Backend**: A secure, scalable backend for managing orders, payments, and real-time updates, ensuring smooth operations even during peak times.

Timeline:

The development of the QR Code Menu System was structured across 17 weeks, following a clearly defined schedule from research and design to final deployment and documentation. This section details each week's activities, ensuring clarity on the progress and workflow of the entire project.

Week 1–2: Research & Planning

- **Define Objectives:**
  Clarified the main aim of the project—to modernize the food ordering experience using QR codes—and set specific goals such as reducing wait times, improving accuracy, and enhancing convenience.
- **Literature Review:**
  Studied academic papers and technical articles related to digital food ordering systems, QR code technologies, and UI/UX best practices to build a theoretical foundation.
- **Problem Analysis:**
  Identified key pain points in the traditional cafeteria ordering process, such as long queues, communication errors, and lack of order tracking.
- **Competitor Review:**
  Analyzed local and international systems (e.g., Domino's QR menu, Uncle Hai) to understand existing solutions, uncover limitations, and spot opportunities for innovation.

Week 3: Design & Prototyping

- **UI/UX Wireframes:**
  Designed wireframes for both customer and vendor views, focusing on intuitive layout, minimal steps, and mobile responsiveness.
- **Logo & Branding Mockups:**
  Created the visual identity for the system, including the "Nhamey" logo, using blue and white colors to reflect freshness and trust.
- **System Architecture Diagram:**
  Mapped out the system structure, showing how the frontend, backend, database, and QR code logic interact within a modular architecture.

Week 4–11: Frontend Development

- **Nuxt.js + Vue.js Interface:**
  Built a responsive single-page application (SPA) that loads menus, handles user interactions, and shows order statuses.
- **Tailwind CSS Styling:**
  Applied Tailwind CSS to ensure consistent, modern styling with a mobile-first approach.
- **ShadCN UI Components:**
  Used open-source UI components from ShadCN to accelerate development and maintain design consistency.
- **Real-Time Menu Feature:**
  Implemented a dynamic menu that updates in real-time, reflecting item availability and vendor changes without reloading the page.

Week 4–11: Backend Development

- **Express.js APIs:**
  Developed RESTful APIs for actions such as placing orders, viewing menus, and managing tables.

- **MongoDB Atlas:**
  Used MongoDB Atlas as the cloud-hosted database to store food items, order details, user information, and table links.
- **JWT Authentication:**
  Secured user sessions with JSON Web Tokens (JWT) to prevent unauthorized access and protect vendor operations.
- **QR Code Generation:**
  Automatically generated unique QR codes for each table, linking them to specific table IDs for accurate order mapping.
- **Order Tracking Logic:**
  Built logic to track order status (pending, preparing, completed) and display updates in real-time for users and vendors.

Week 12–13: Integration & Testing

- **Frontend-Backend Integration:**
  Connected APIs to the frontend interface and verified real-time updates and data flow.
- **Postman Testing:**
  Used Postman to test endpoints individually, ensuring all API responses were accurate and secure.
- **Swagger Documentation:**
  Created API documentation using Swagger UI, allowing developers and testers to interact with the backend easily.
- **Bug Fixing & Optimization:**
  Identified and resolved issues related to login flow, order delays, UI glitches, and data synchronization.

Week 14: Deployment

- **Frontend Deployment (Vercel):**
  Deployed the frontend codebase to Vercel for fast, globally distributed access with Git-based versioning.

- **Backend Deployment (Render):**
  Deployed Express.js APIs to Render, configured environment variables, and ensured stable runtime.
- **MongoDB Atlas Connection:**
  Integrated database with the deployed backend to ensure real-time and persistent data access.
- **CI/CD Setup:**
  Set up Continuous Integration/Continuous Deployment pipelines to automate future updates from GitHub to the live system.

Week 15: Survey & Feedback

- **User Testing:**
  Invited students, cafeteria vendors, and staff to test the system in a live environment.
- **Feedback Collection:**
  Distributed surveys and conducted interviews to assess ease of use, response time, and clarity of interface.
- **Common Concerns Identified:**
  Users appreciated the concept but raised concerns about internet dependency and possible confusion during first-time use.

Week 16–17: Report & Documentation

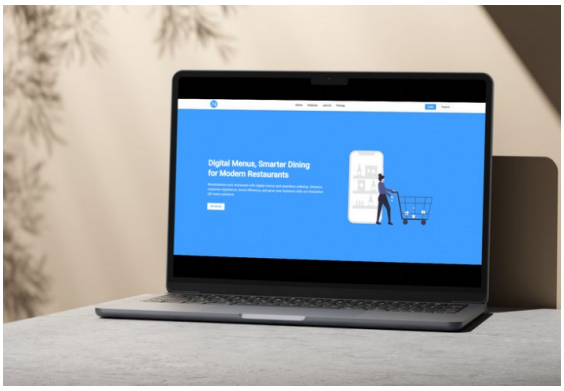
- **Finalize Written Report:**
  Compiled all findings, system design, implementation details, and test results into the final thesis document.
- **Screenshots & Diagrams:**
  Included architecture diagrams, flowcharts, system screenshots, and wireframes to support the written explanations.
- **Documentation for Future Use:**
  Prepared technical documentation including system setup, deployment steps, and developer notes for potential future improvements or handovers.

Budget:

- **Payment Gateways**: Allocate resources for integrating secure payment systems.
- **Hosting**: Budget for web hosting and database storage (AWS, MongoDB Atlas, etc.).
- **Development Tools**: Costs for necessary development tools and APIs

5.2 Frontend Development

- **Nuxt.js Development** :The frontend was built using **Nuxt.js** for its server-side rendering (SSR) and modular architecture, ensuring fast performance and SEO benefits. Pages, components, and layouts were designed for reusability and consistency. **Dynamic routing** was implemented for menu and order pages, allowing seamless navigation and content updates based on user interactions.
- **Tailwind CSS** was used to create a responsive and clean user interface. Utility classes simplified the styling process, ensuring consistent spacing, alignment, and responsiveness across devices.
- **ShadCN UI** were utilized to accelerate the development of complex elements like modals, forms, and dropdowns, offering customization and accessibility.
- **Vuex** was used to manage application state effectively, handling key data such as user sessions, menu details, and QR code information.
- **Dynamic QR codes** were displayed using backend data, enabling real-time updates and smooth integration into the user interface.



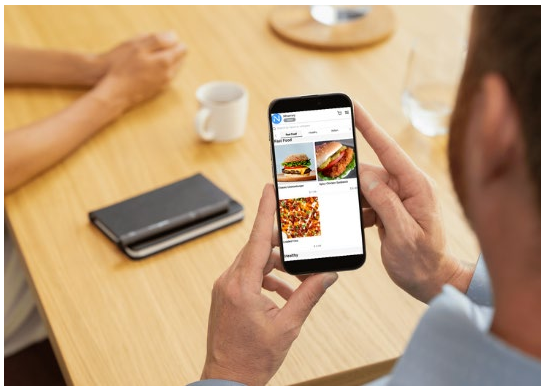
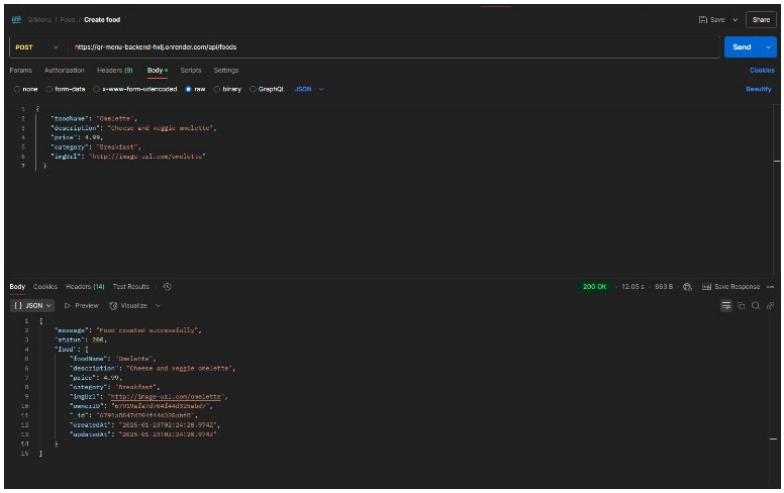


Figure 5: Frontend Mock up design

5.3 Backend Development

- The backend was developed using **Express.js** and exposed a set of **RESTful API** endpoints for managing data. These endpoints supported **CRUD operations** for menu items, orders, and user profiles. The API followed **REST principles** to ensure modularity, scalability, and easy integration with the frontend.
- The **MongoDB** database was structured to store collections for **users**, **menu items**, **orders**, and **QR code metadata**. Indexing was implemented on key fields to optimize query performance, particularly for frequently accessed data like user profiles and menu items.
- User authentication was implemented using **JSON Web Tokens (JWT)** for secure and stateless authentication. **bcrypt** was used to hash and encrypt passwords before storage, ensuring safe transmission and secure storage of user credentials.
- The **qrcode npm package** was integrated into the backend to dynamically generate QR codes for menu items or orders. Customization options, such as size and format, were applied to ensure the QR codes met design and functional requirements.

- API endpoints were rigorously tested using **Postman**, covering various scenarios including edge cases, error handling, and performance under load. This ensured that the API was robust, reliable, and capable of handling real-world usage.



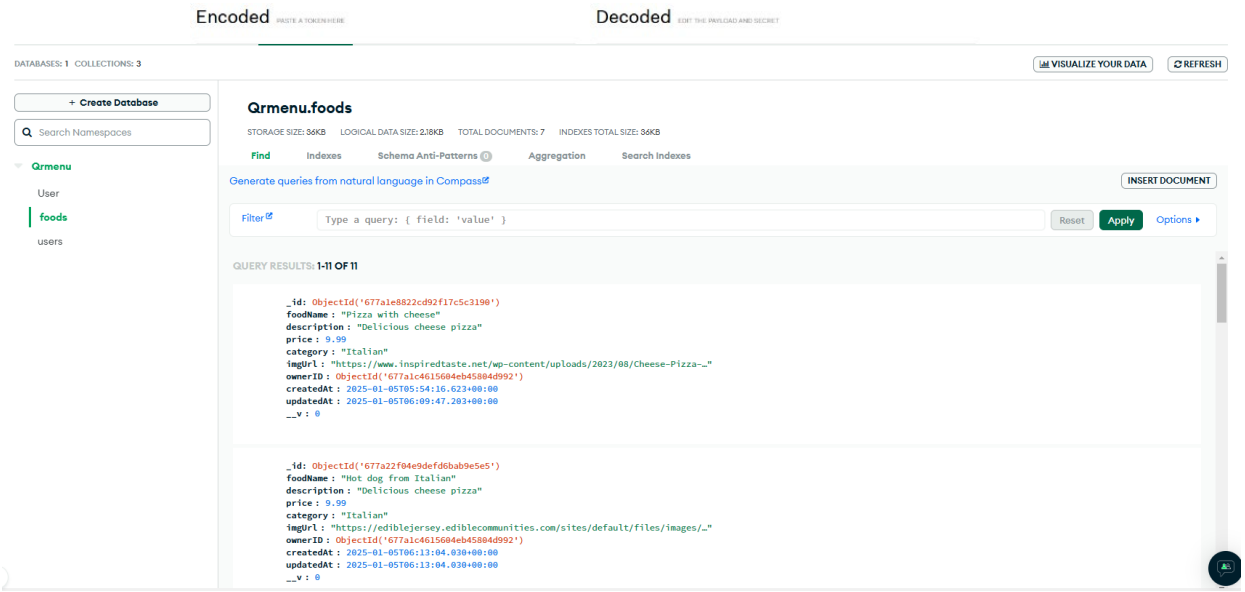


Figure 6: Backend Design

5.3.1 System Architecture Diagram

The system architecture of the **RUPP QR-Menu System** is designed to provide a seamless and efficient digital ordering experience for students and staff while ensuring secure,

scalable, and modular backend services. The architecture consists of the following core components:

1. **QR Code Web Portal**
   Users (customers or staff) access the system by scanning a QR code which directs them to the web portal. This portal serves as the entry point to the menu and ordering platform.
2. **Client Interfaces**
   - **Customer Interface**: Displays the digital menu and allows users to place orders, view order status, and receive notifications.
   - **Staff Interface**: Enables staff to manage menus, view and process orders, generate reports, and track payment statuses.
3. **Authorization Layer**
   Handles user login and access control using secure authentication mechanisms. Staff access is validated before interacting with the system.
4. **Backend (REST API with Microservices)**
   The backend is composed of multiple microservices, each dedicated to a specific function:
   - **Authentication Service**
   - **Menu Management Service**
   - **Order Processing Service**
   - **QR Code Service**
   - **Report Service**
   - **Payment Receipt Service**

   These services communicate via REST APIs and handle the business logic required for the system's functionality.
5. **Cloud-Based Database Platform (MongoDB Atlas)**
   MongoDB Atlas is used to store persistent data such as user profiles, food menus, order histories, and reports. It ensures high availability and cloud scalability.
6. **Notification System**
   Sends real-time updates to customers and staff (e.g., order confirmation, preparation status, or payment receipt) to keep all stakeholders informed.

7. **Server & Hosting Infrastructure**

The backend microservices are hosted on a secure server environment, and MongoDB Atlas is integrated to ensure cloud data access. The entire system is deployed on reliable cloud platforms (e.g., Vercel and Render).

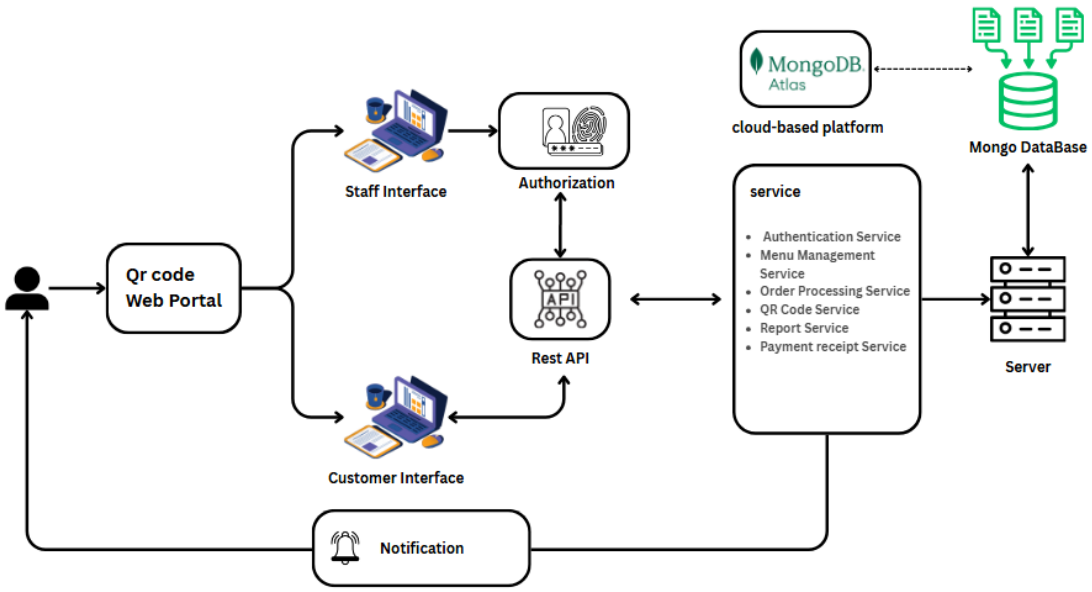


Figure 7: System Architecture Diagram

5.4 Deployment Process

The backend was deployed on **Render**, which provided a simple and scalable environment for deploying server-side applications. Render allowed for automatic scaling, easy configuration of environment variables, and smooth integration with the MongoDB database.

The frontend was hosted on **Vercel**, which offered a globally distributed CDN for fast content delivery. Vercel also enabled automated build processes and seamless integration with GitHub, ensuring continuous deployment with minimal manual intervention.

The database was hosted on **MongoDB Atlas**, offering a secure, scalable, and reliable environment for data storage. MongoDB Atlas provided features like automated backups, real-time monitoring, and high availability to ensure data integrity and reliability.

The project's code was stored on **GitHub**, providing version control, collaborative development, and easy integration with the frontend and backend deployment services.

Figure 8:Hosting and Storing technology

5.5 User's Use Cases

Use Case Name: ***Food Ordering via QR-Menu System (Nhamey)***

Actors:

• Customer (Students, Staff): Scans QR codes, places orders, makes payments, and views order status.

• System Administrator: Ensures system functionality and performance.

Description:

The RUPP QR-Menu system enables students and staff to scan a QR code, access the cafeteria menu digitally, place orders, and complete payments through a mobile-friendly platform. It streamlines the ordering process and enhances the dining experience within the Royal University of Phnom Penh (RUPP).

Preconditions:

• The customer has a smartphone or laptop with internet and a QR scanner.

• The system is deployed and accessible via QR code or link.

• Menus are up-to-date and published by cafeteria vendors.

• Stable internet connection is available on campus.

Basic Flow:

1. **Customer Scans QR Code (User Device)**
   • Device: Smartphone or Laptop
   • Action: Customer scans a QR code located at a table or cafeteria entry point.
   • Redirect: System opens the Digital Menu showing food items, prices, and images.
2. **Place Order (Frontend)**
   • Action:
   o Customer browses the menu and selects desired items.
   o Customizes food options if needed (e.g., spicy level, no meat).
   o Chooses dining preference: Pickup or Dine-in.
   o Proceeds to checkout.
   • Payment: Completes payment via available method (e.g., cash, mobile payment).
3. **Order Processing (Backend / Database)**
   • Storage: Order details saved to MongoDB.
   • Admin Dashboard: Vendor receives real-time order notification.
   • Status Update: Order status updated as Preparing → Ready.
4. **Order Ready Notification (back office)**
   • Customer can view real-time status on their screen.
   • System notifies when order is ready for pickup or dine-in serving.
5. **Completion**
   • Customer collects the food.
   • System marks the order as complete.
   • Optionally, customer can enjoy.

Postconditions:

• Orders are fulfilled and marked complete.

• Payments are successfully processed.

• Optional customer feedback is recorded for future vendor insights.

Exceptions:

• No Internet Connectivity: Customer is prompted to reconnect.

• Payment Failure: User is prompted to retry or switch method.

• Menu Load Error: Displays error message and fallback option.

Frequency of Use:

• Daily, especially during lunch and dinner times.

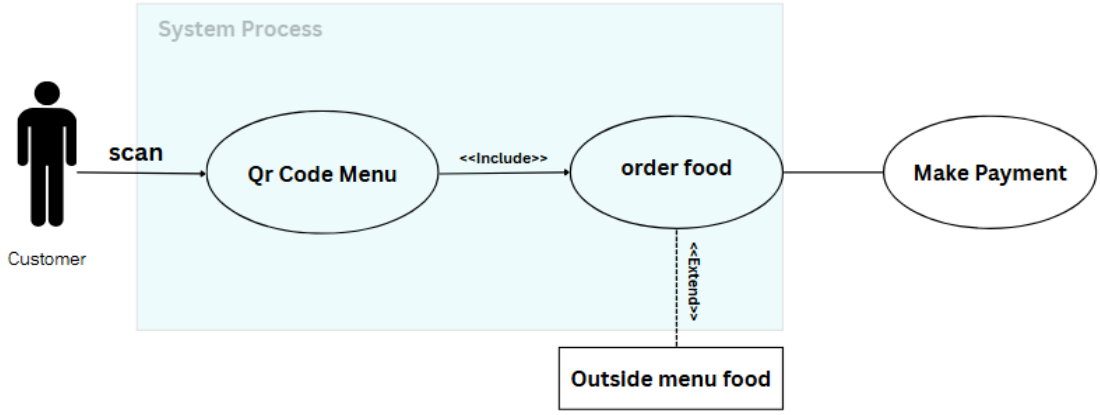


Figure 9: Customer use case

Use Case Name: Vendor Menu and Order Management via QR-Menu System (Nhamey)

Actors:

• Cafeteria Staff/Vendor: Manages food listings, receives and processes orders.

• System Administrator: Maintains backend system, ensures uptime and performance.

Description:

This use case allows vendors to manage their menu, monitor real-time orders, update order statuses, and maintain service efficiency through the admin dashboard.

Preconditions:

• Vendor is registered and authenticated in the system.

• Internet connection is stable.

• Menu items are already created or ready to be added.

Basic Flow:

1. **Login and Dashboard Access**
   • Vendor logs into the system using secure credentials.
   • Dashboard loads showing overview of food items, orders, and recent activity.
2. **Menu Management**
   • Vendor creates, updates, or deletes food items.
   • Adds images, descriptions, prices, and availability for each menu item.
   • Changes appear in real time to customers who scan the QR code.
3. **Order Monitoring and Processing**
   • New orders appear in real time on the dashboard.
   • Vendor views order details and begins preparation.
   • Vendor updates status: Received → Preparing → Ready.
   • System auto-updates the customer interface with real-time status.
4. **Finalization**
   • Once food is served, the order is marked as completed.
   • Order history is saved for review and reporting.

Postconditions:

• Orders are completed and archived.

• Menu remains current and accurate.

• Operational data is updated and synced with the system.

Exceptions:

• Internet/Connection Issues: Vendor prompted to reconnect.

• Input Errors: System validation flags missing fields or incorrect data.

• System Downtime: Fallback offline protocol or order tracking via manual list.

Frequency of Use:

• Daily, particularly before meal hours and during service peaks.

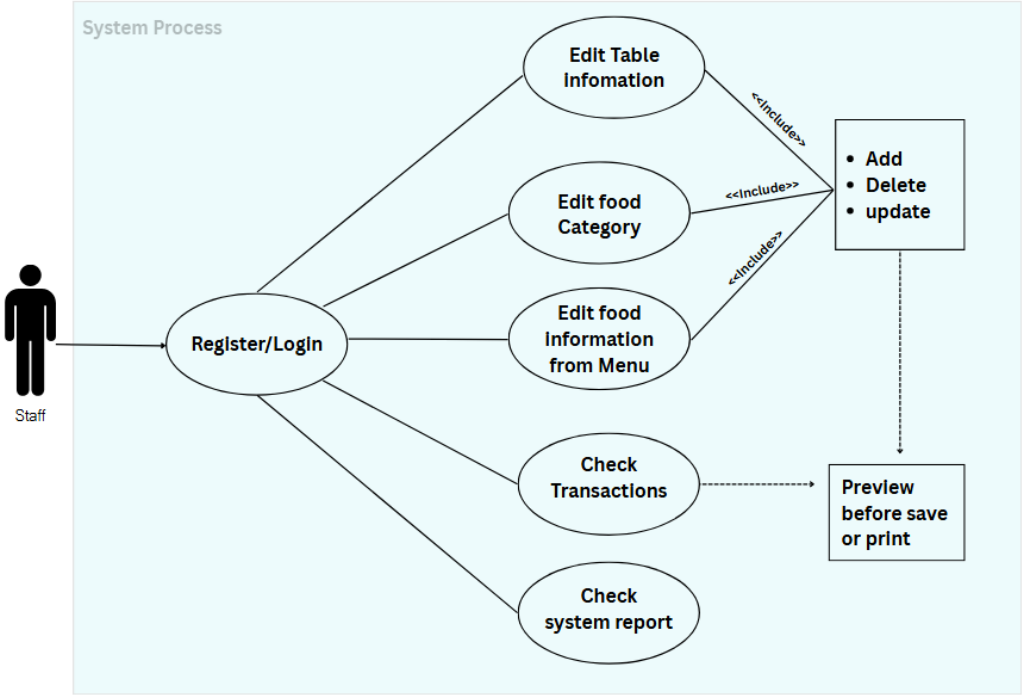


Figure 10: Staff use case

5.3 Full System Flow Description: Customer and Staff Interaction

This flow diagram illustrates the complete operational cycle of the QR Code Menu System, beginning from the customer's interaction with the platform and ending with the fulfillment of the order. The process involves two main roles: the **Client (Customer)** and the **Admin Staff**, with supporting systems such as the **Menu Interface**, **Database**, and **Admin Dashboard** ensuring data processing and service coordination.

1. Client Access via User Device

The journey starts with the **client (customer)** using a **smartphone or laptop** to engage with the system. By either scanning a QR code or inputting a provided link, the user is directed to the menu interface of the system.

2. Accessing the Menu Page and Placing an Order

Once redirected, the customer views the **menu page**, where they can browse available food items and place an order directly through the web interface. This stage takes place entirely within the **QR Menu module**, where the frontend handles user interactions.

3. Order Request and Database Validation

After the customer submits the order, it is sent to the **MongoDB database** through backend processing. The system checks the data for validity and stores the order record. If successful, the flow continues to the next stage. If unsuccessful, the customer is notified and prompted to return to the menu or retry the process.

4. Order Confirmation and Cash Payment Handling

If the order is successfully stored in the database, the system may then initiate the **cash payment process** (depending on configuration). Although payment by cash is handled offline, the order status is updated in the system to reflect that it's awaiting payment or has been paid, depending on when the transaction occurs.

5. Admin Dashboard Monitoring

In the **back office**, staff use the **Admin Dashboard** to monitor all new and pending orders in real time. They can view order details, update statuses, and manage operational flow based on the incoming data.

6. Receipt Generation and Order Processing

Using the admin dashboard, staff can **generate an order receipt**. This receipt contains key information such as order ID, food items, total cost, and table reference. It may also serve as a reference for cooking, serving, or customer verification.

7. Final Steps and Completion

Once the order is confirmed, paid for, and the receipt is generated, the process enters its final phase. Food is prepared and delivered or served accordingly. The system may optionally

capture the final payment confirmation (if post-paid) and update the record in the database. With all steps complete, the order lifecycle ends successfully.

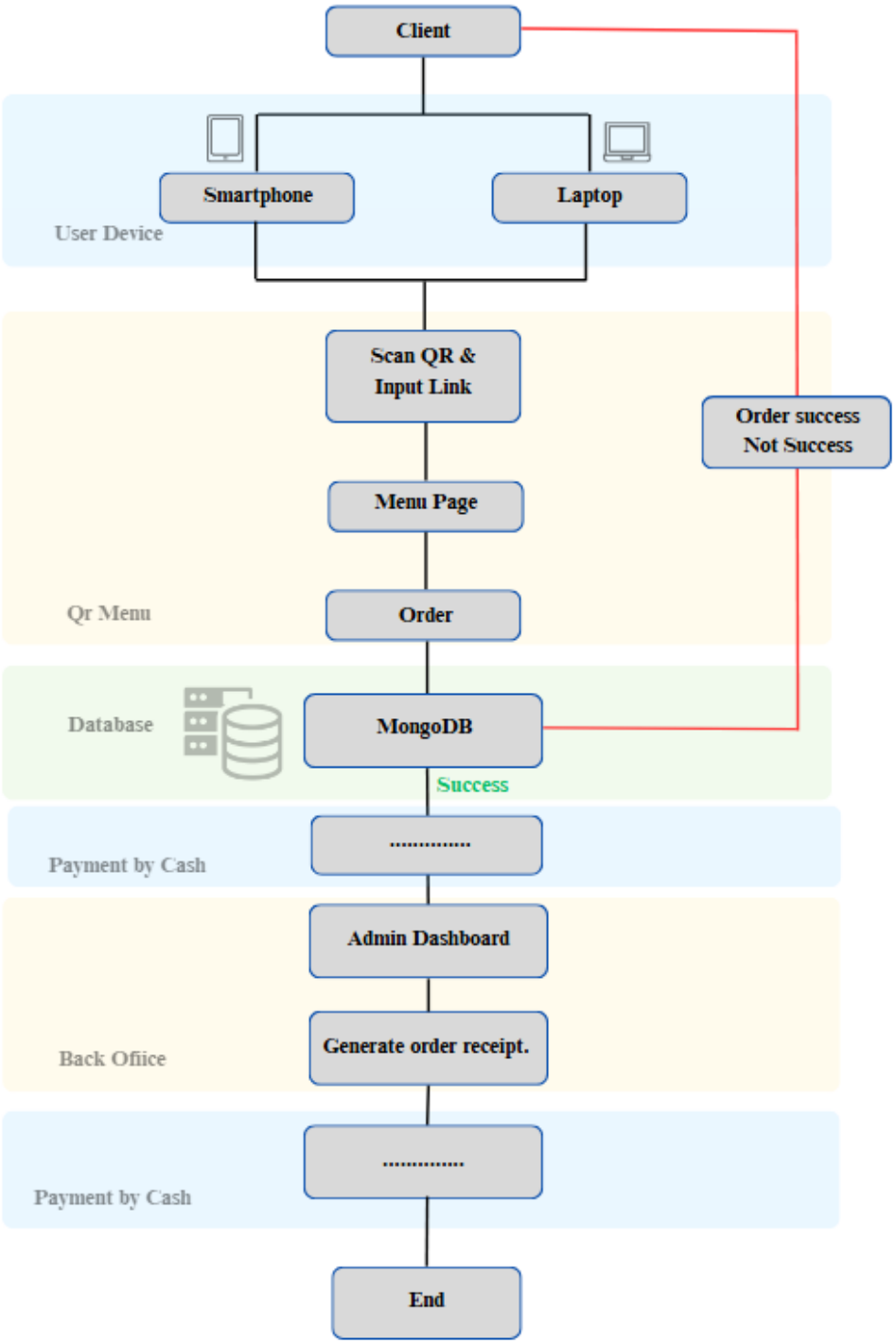


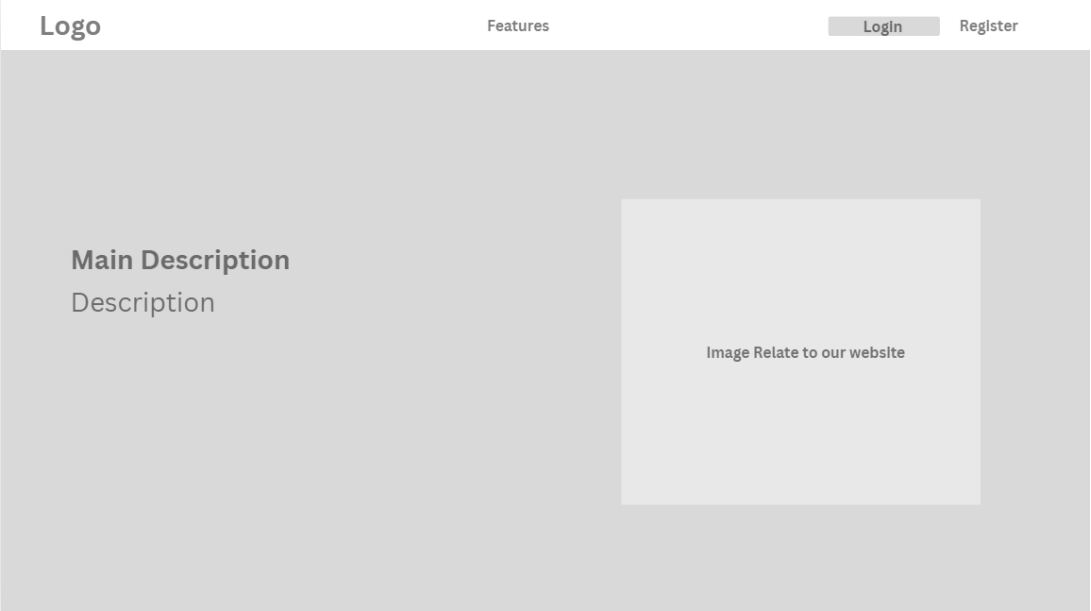
Figure 11: Full System Flow

CHAPTER 6 : UX/UI MOCK UP and RESULTS

6.1 UX/UI Mock up

This mock-up section showcases the "NHAMEY" logo applied across various elements, created using Photoshop, Canva, and mockup resources from mockupworld.co for a comprehensive UX/UI design. The process began in Photoshop, where the logo's blue (#39a0ff) and white (#ffffff) color scheme was refined to reflect the food-inspired theme, ensuring a clean and vibrant aesthetic. Canva was then utilized to craft the initial layouts, allowing for easy adjustments and alignment with the brand's fresh, ocean-evoked identity. Finally, mockupworld.co provided high-quality templates, including the apron in a kitchen setting, a storefront sign on a brick building, and a takeout bag on an orange surface. These templates enabled realistic placement of the logo, demonstrating its versatility for a seafood restaurant's staff uniforms, exterior branding, and packaging. The combination of these tools ensured a polished UX/UI mock-up, effectively highlighting the logo's role in creating an appetizing and cohesive food dining experience.

6.1.1 Main page



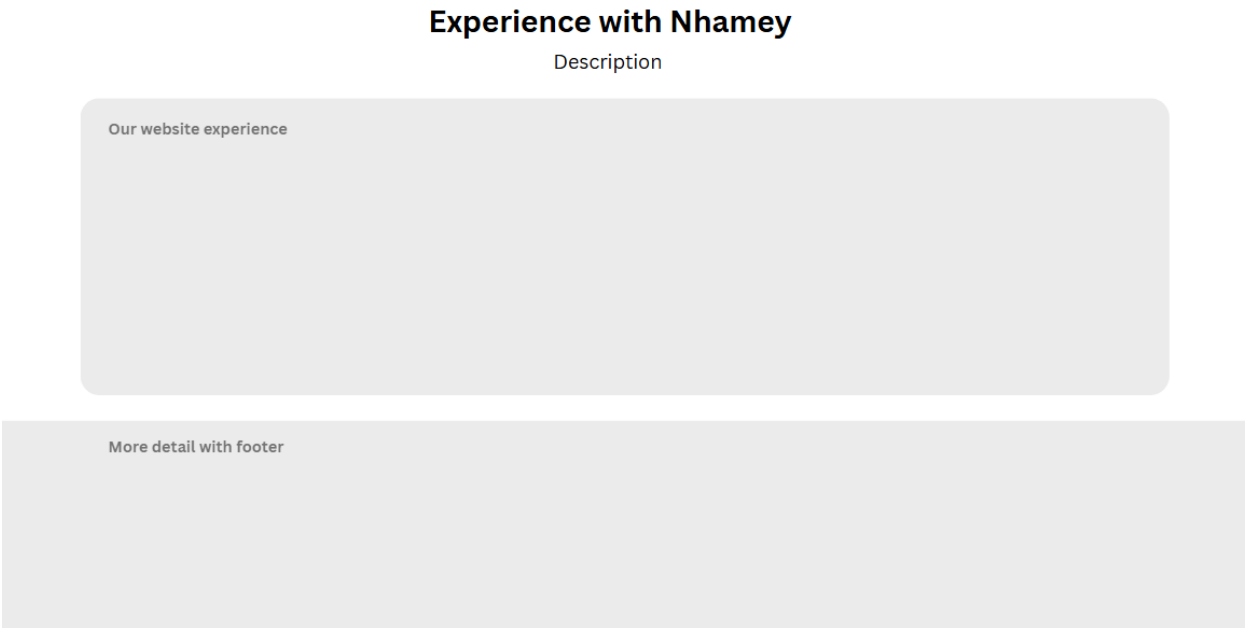


Figure 12: Mock up main page

6.1.2 Dashboard page

Figure 13: Mock up Dashboard

6.1.3 Menu page

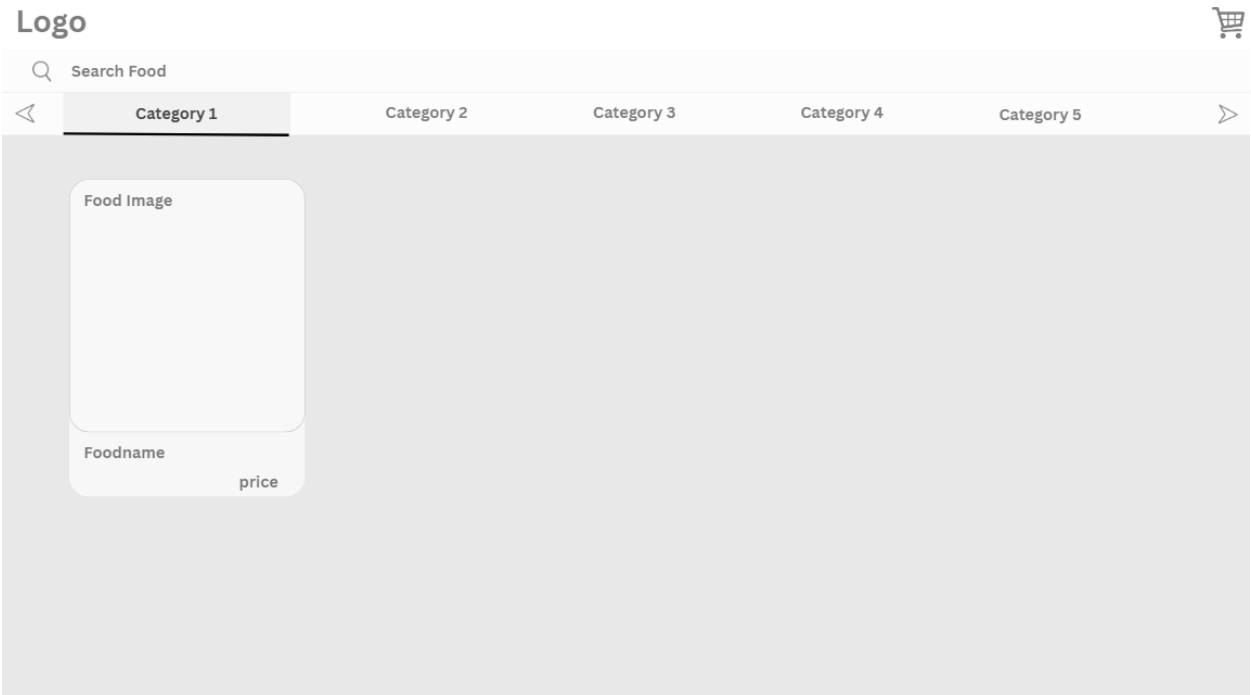


Figure 14: Mock up menu page

6.1.4 Login & Register page

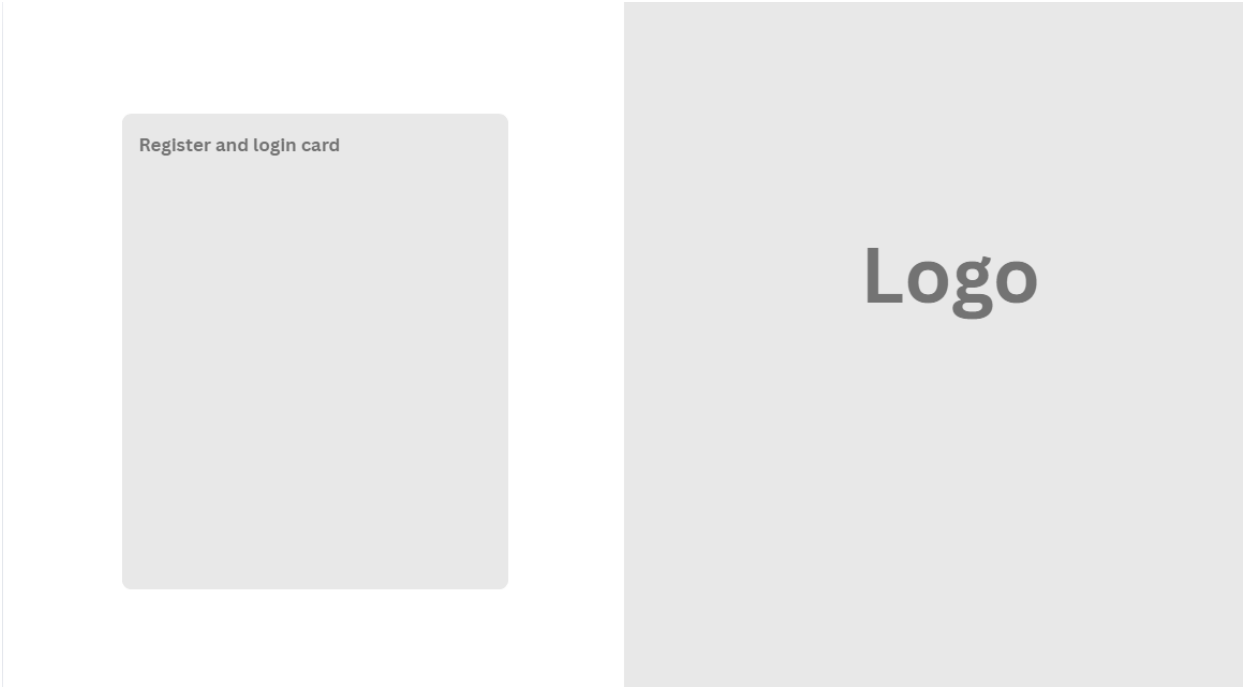


Figure 15: Mock up authorization page

6.1.5 Logo Design and Mock Up

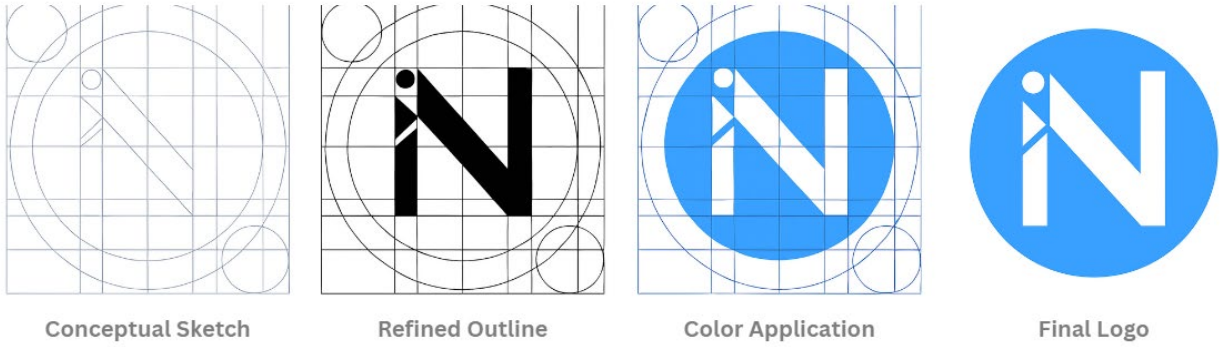


Figure 16: Logo Design

- **Conceptual Sketch**: A basic outline with geometric grids and circles, showing the initial "N" and dot design, inspired by a human who stand order system .
- **Refined Outline**: A polished black "N" with the dot, refined on a grid for symmetry, symbolizing the structured preparation of a fresh food dish.
- **Color Application**: The design gains a blue (#39a0ff) circular background, evoking the ocean's freshness, with a white (#ffffff) "N" and dot for purity, mirroring clean food presentation.
- **Final Logo**: The completed logo without guidelines, showcasing the blue and white color scheme, ready to represent a fresh, appetizing seafood brand.

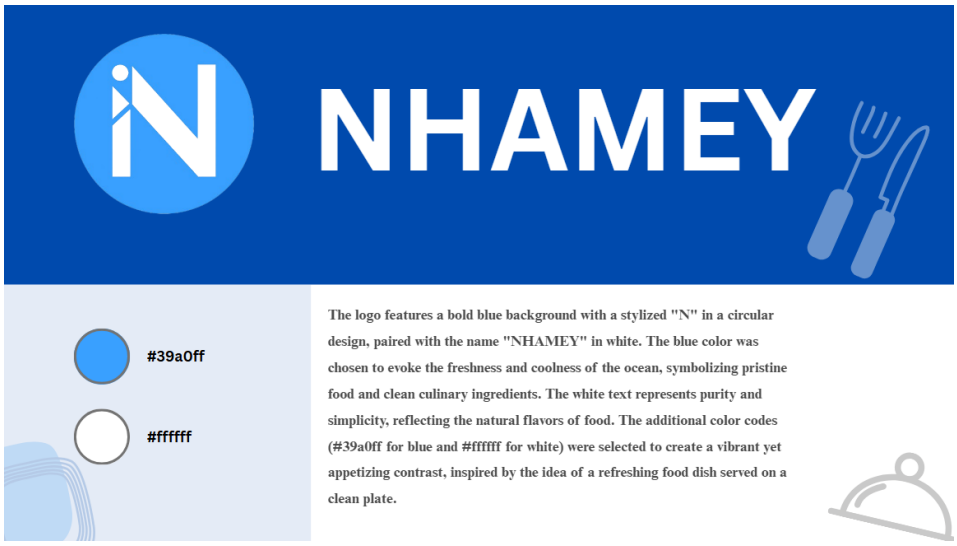


Figure 17: Logo detail

This section showcases the final "NHAMEY" logo and its description, emphasizing its food-inspired design. The logo features a bold blue (#39a0ff) background with a stylized white (#ffffff) "N" in a circular design, paired with the name "NHAMEY" in white.

Figure 18: Brand mock up

This mock-up section displays the "**NHAMEY**" logo on an apron in a kitchen, a storefront sign, and a takeout bag, showcasing its food-inspired design with blue and white colors for freshness and purity.

6.2 Result

At this stage, we have successfully completed the design and development of our website. This includes the creation of a professional logo and branding that align with the identity and vision of our platform. The UX/UI mockup was designed to prioritize user experience, clarity, and responsiveness, serving as a guide for development. Following the mockup, we fully built the website based on the approved design. The final product reflects a consistent visual identity, smooth navigation, and functional features as planned, ready for testing and deployment.

6.2.1 Main page

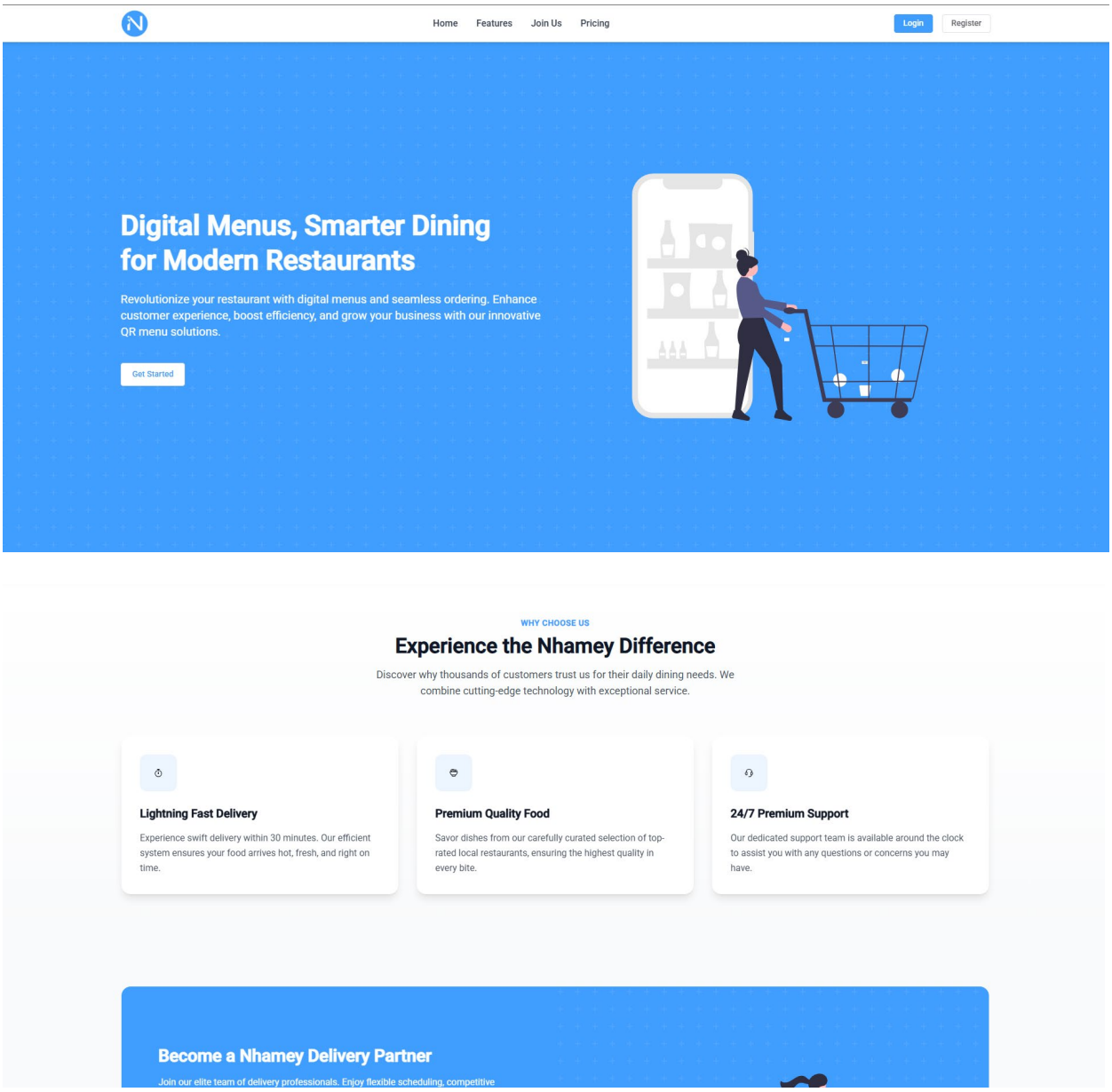


Figure 19: Result of main page

This is our **main page** where cafeteria owners can register or log in to the dashboard to create food items, view details about their food, and get an overview of their sales and orders.

6.2.2 Login and register Page

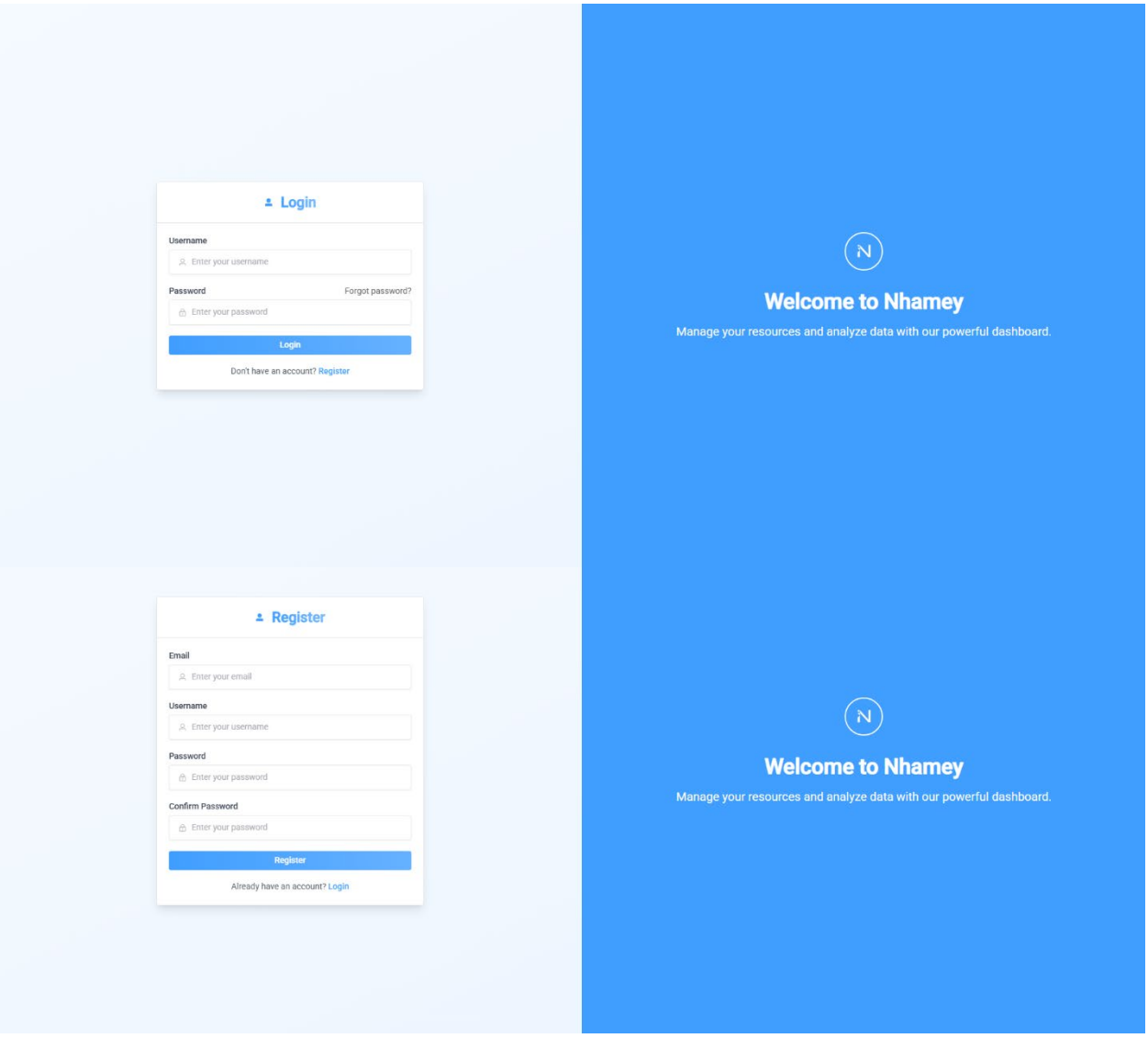


Figure 20: Result of authorization page

This is our login and registration page, where cafeteria owners can securely sign up for a new account or log in to their existing account. Once logged in, they will be redirected to the dashboard, where they can manage their food menu, view detailed information about each food item, track sales performance, and monitor customer orders in real-time.

6.3.3 Dashboard page

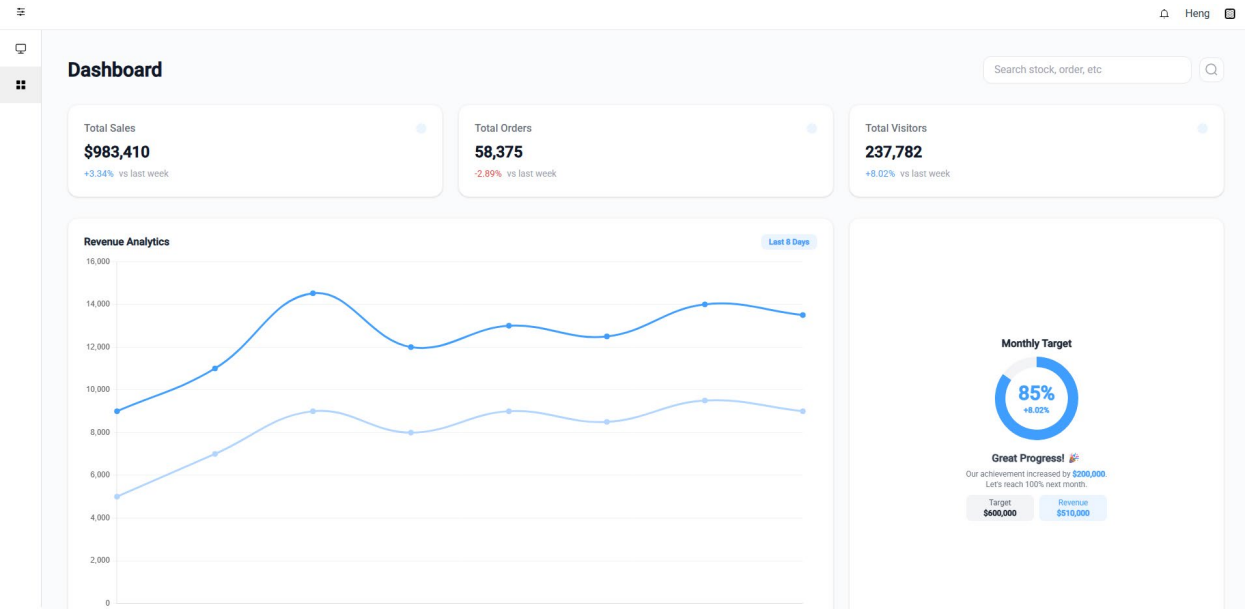


Figure 21: Result of report page

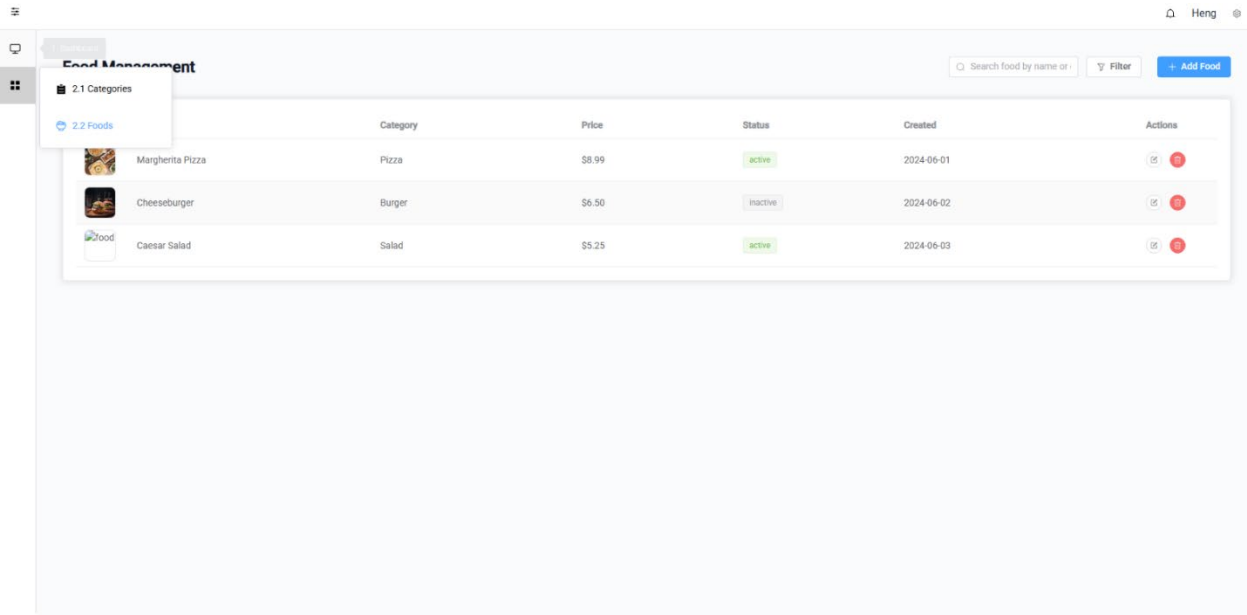


Figure 22: Result of create food page

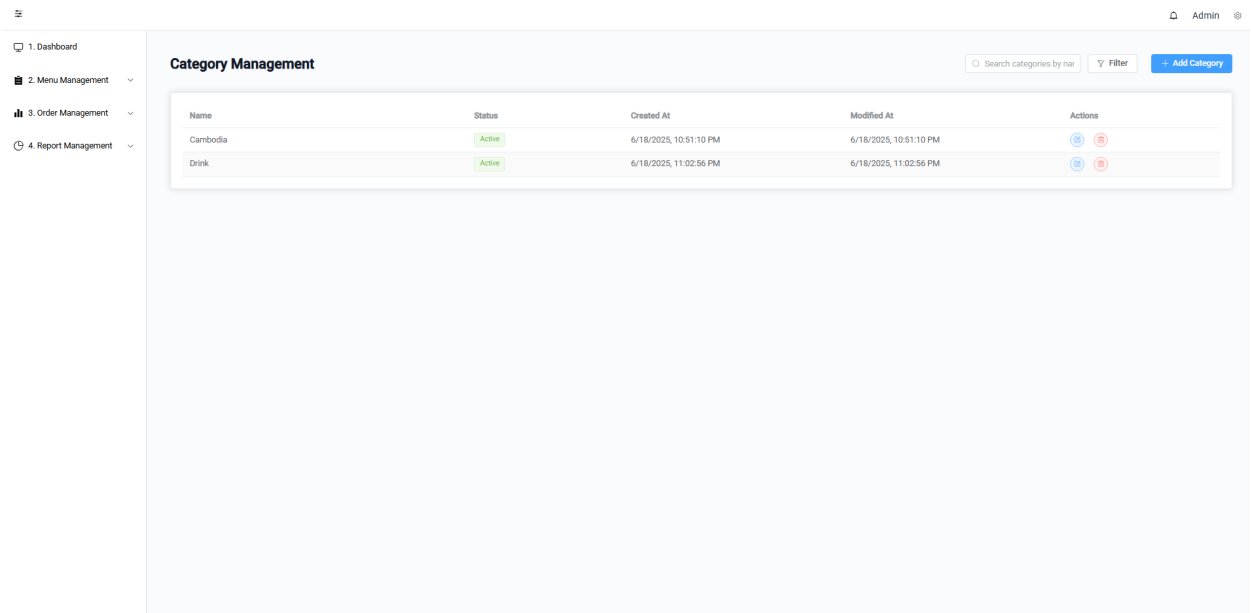


Figure 23: Result of create category Page

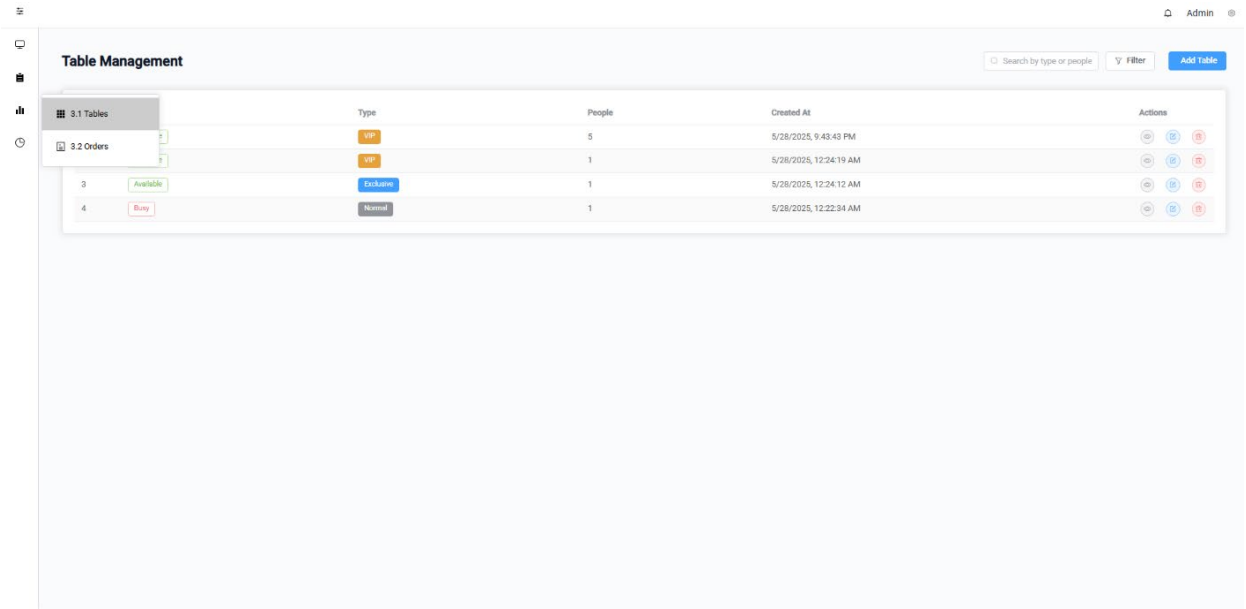


Figure 24: Result of create table page

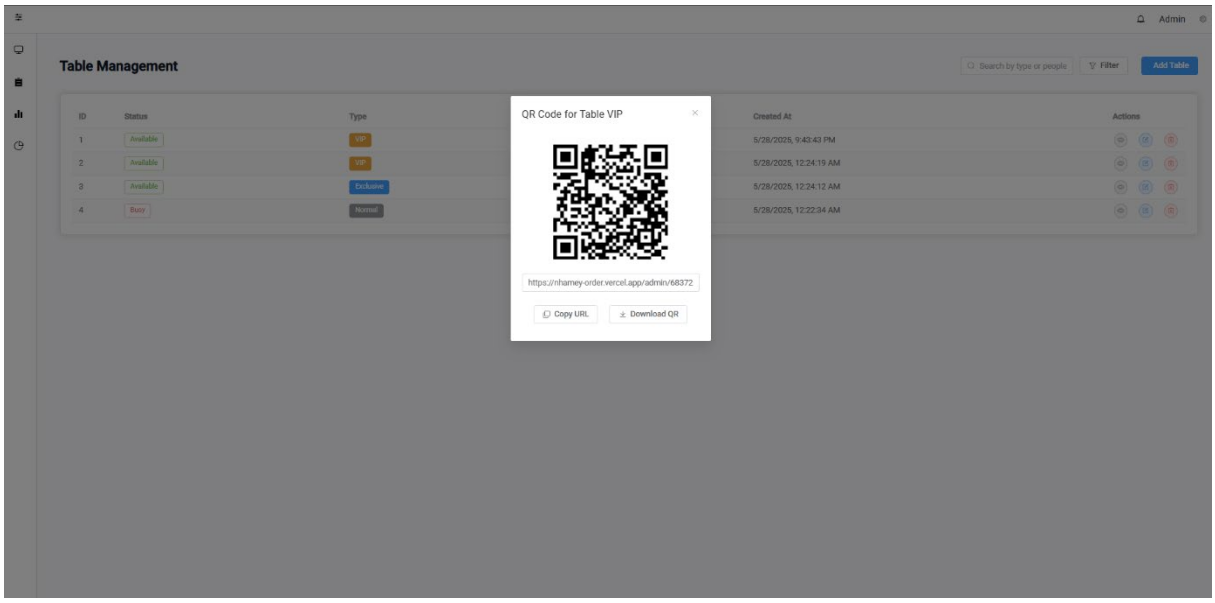


Figure 26 : Result of Table generate QR code

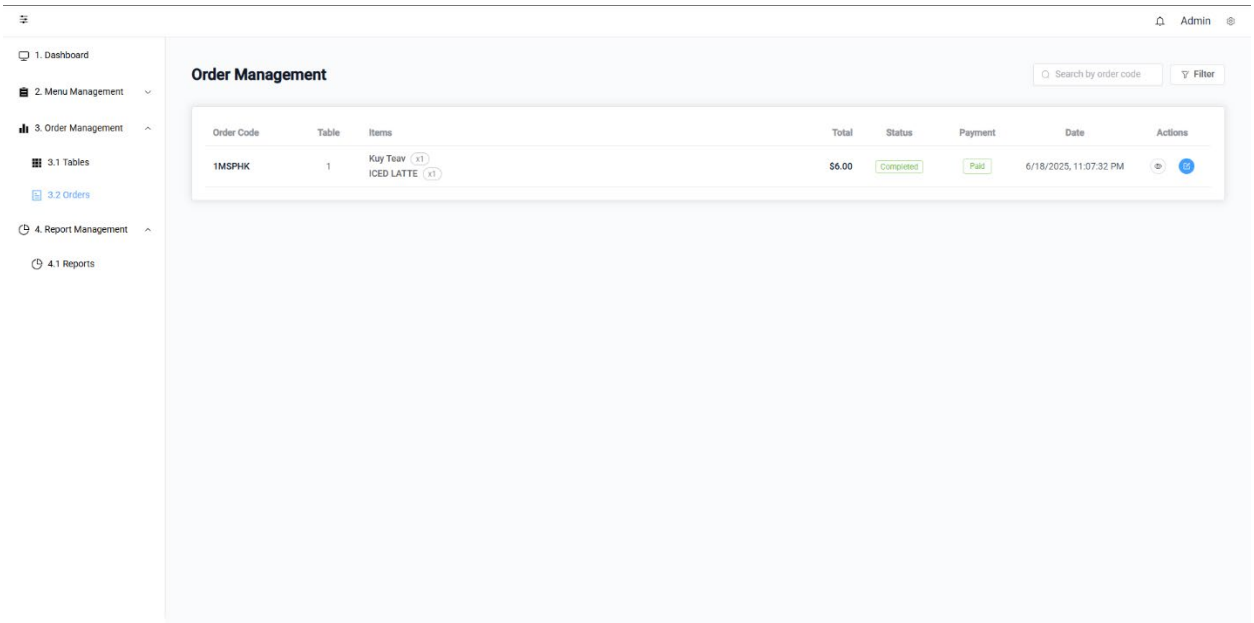


Figure 25: Result of order information

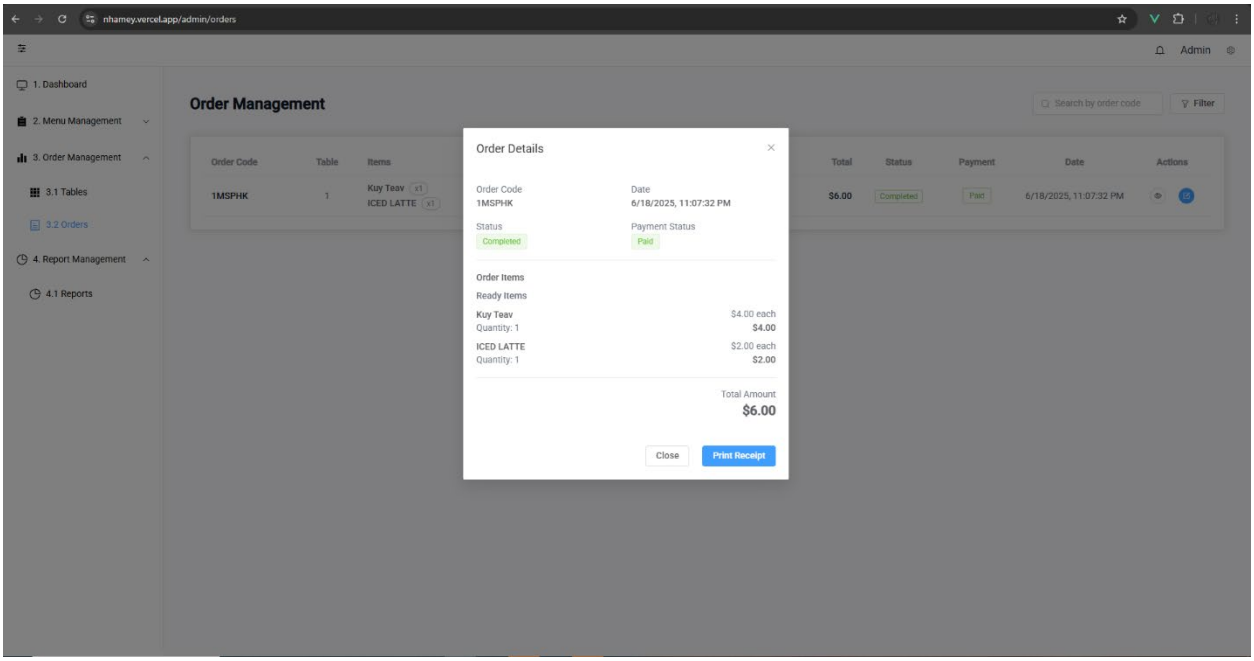


Figure 27: Result of Receipt page

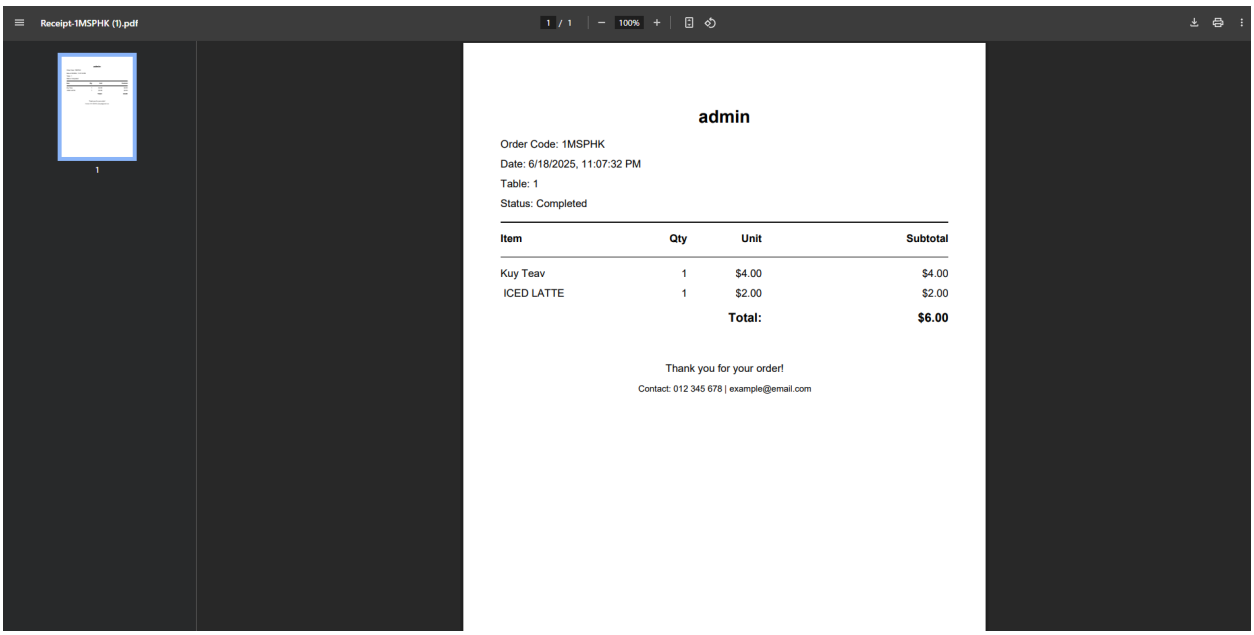
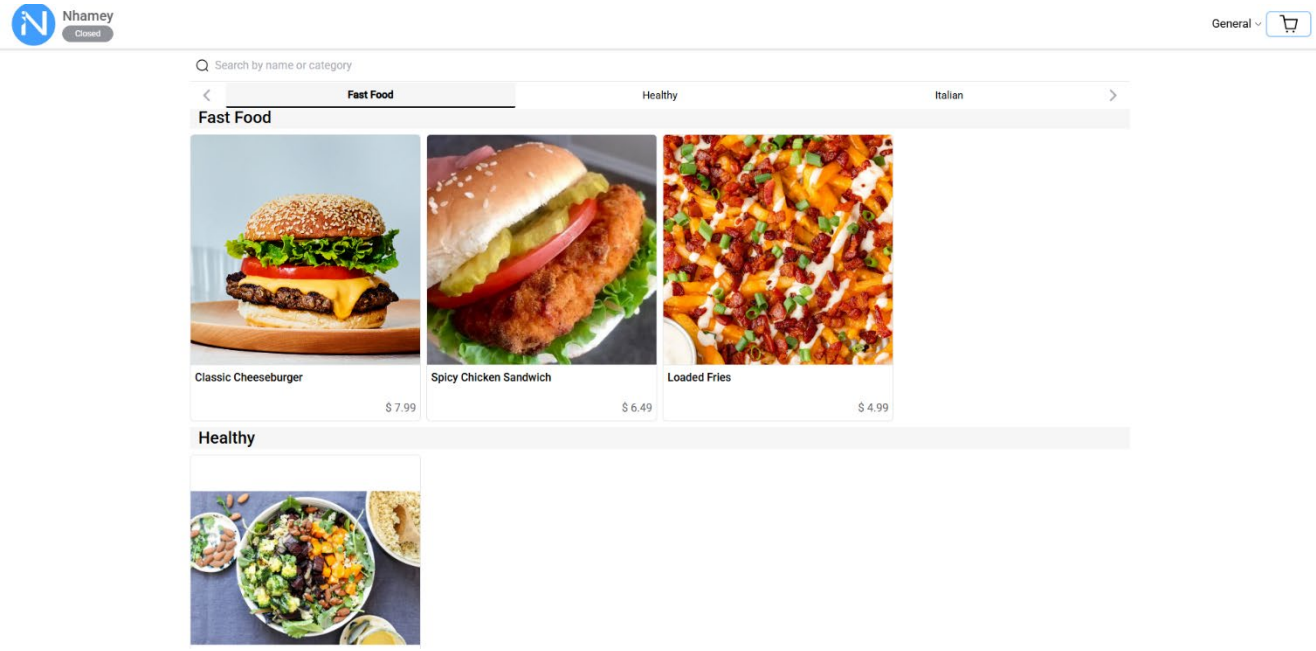


Figure 28 : Result of printing receipt

This is the dashboard, where you can create new categories, add food items, and manage your menu. You can also create tables and generate unique QR codes for each table, allowing customers to place orders directly. Additionally, view detailed information about customer orders, track your sales performance, and monitor overall business activity in real-time.

6.4.4 Menu Page



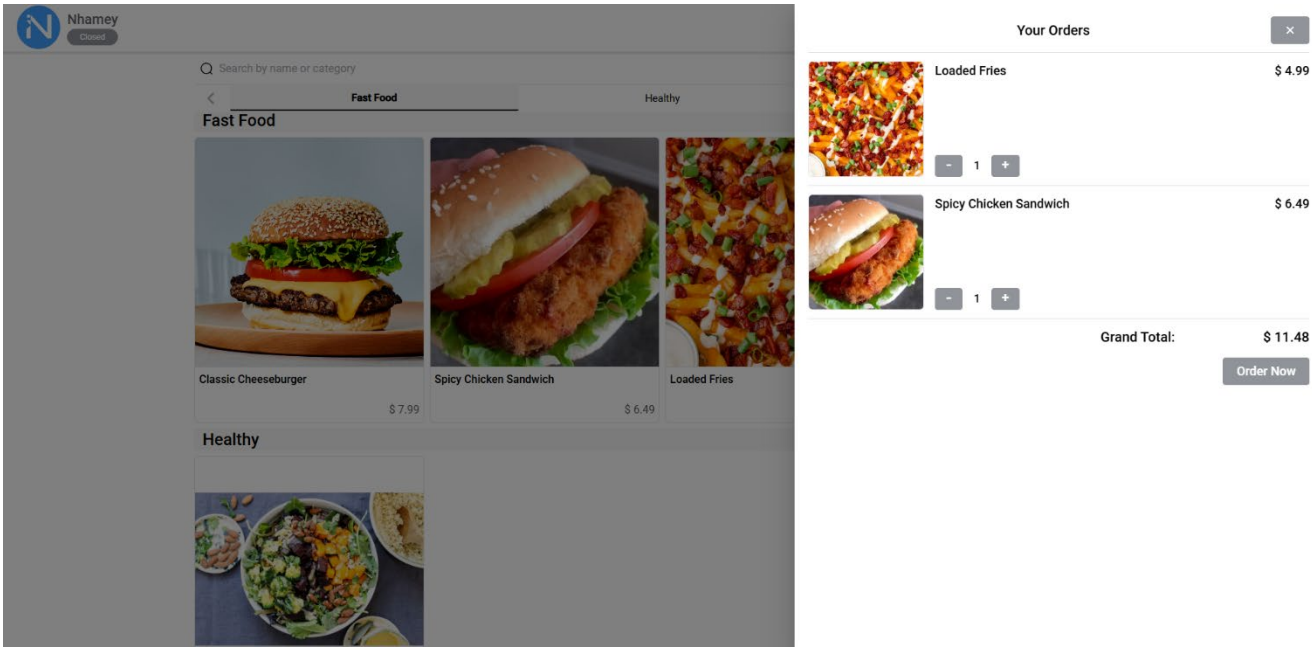


Figure 29: Result of Menu Page

This is our menu page, where customers can browse food items by category, place orders, and make purchases easily. What makes it even more special is the added feature that allows users to switch the language to Khmer for a more localized experience.

6.4.5 Survey of the system

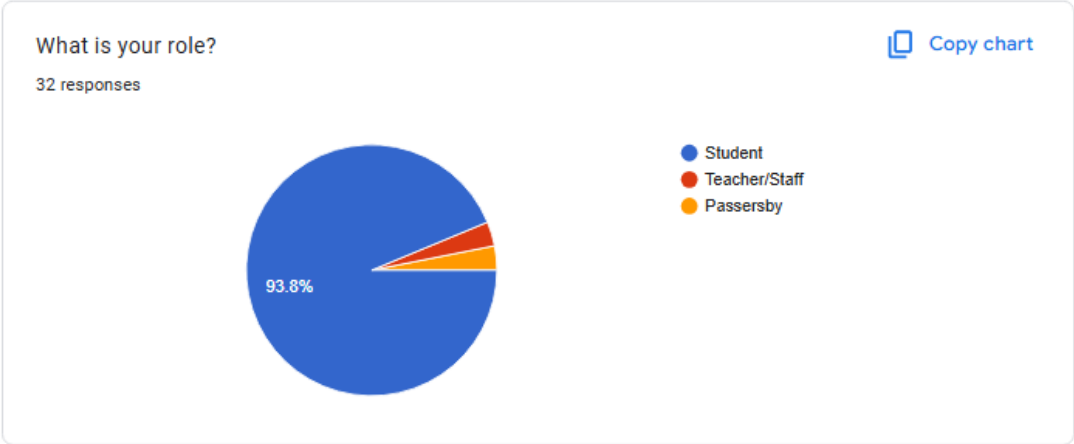
A total of 32 responses were collected in the student survey to assess perceptions of the proposed QR Code Menu System. As reflected in the chart, the majority of respondents (53.1%) were **Year 4** students. This was followed by equal proportions of **Year 2** and **Year 3** students, each contributing 18.8% of responses. A smaller portion of the participants were **Year 1** students and **graduates**.

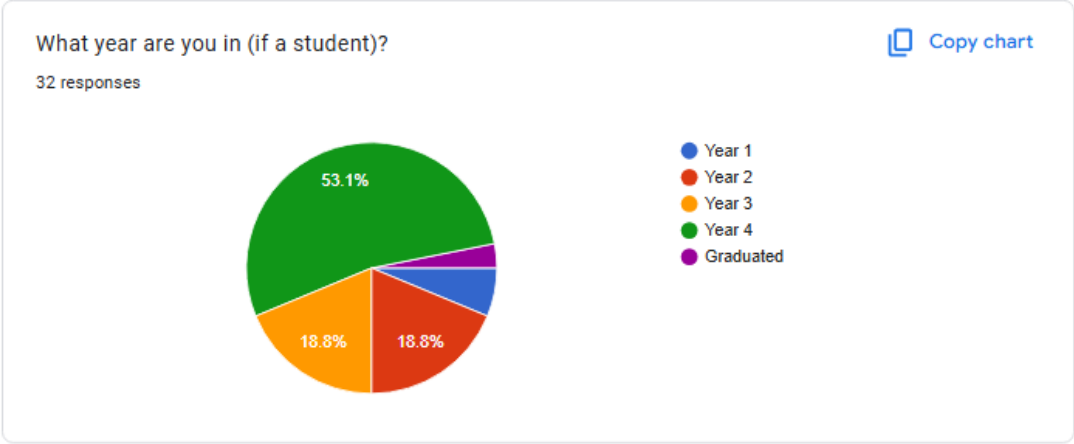
Most participants agreed that the system would be beneficial in enhancing the cafeteria experience. They pointed out key advantages such as reducing waiting times, simplifying the ordering process, and offering a convenient mobile-based solution. The ability to scan a QR code and place orders directly from their own devices was seen as a modern and efficient improvement over traditional methods.

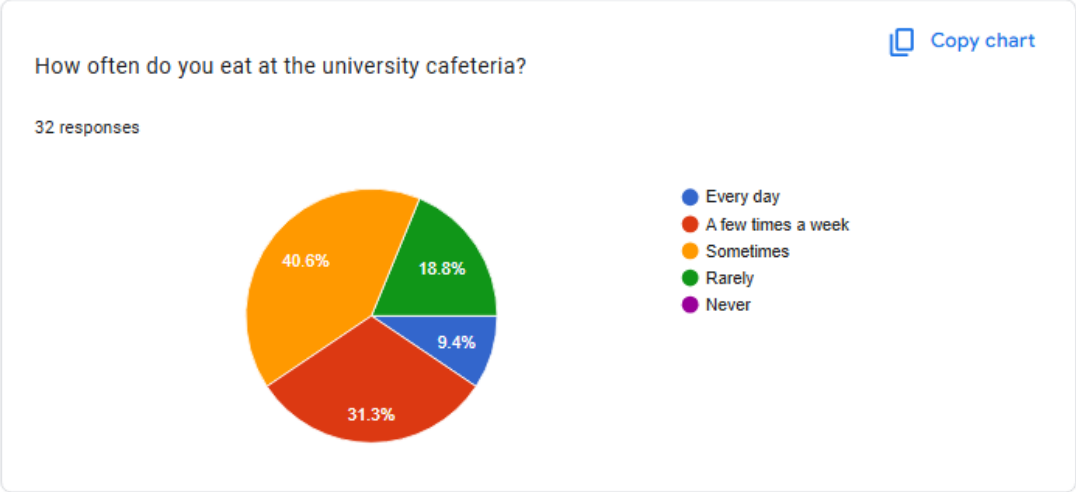
However, several students expressed concerns regarding the **potential performance of the system in low-internet environments**. They noted that if the connection is slow or unstable—especially during busy hours—the system might become difficult to access, leading to delays or

failed transactions. This feedback highlights the need for the system to be optimized for speed and responsiveness, even under limited connectivity conditions.

In addition to this question, the survey included another item aimed at gathering deeper insight into student expectations and concerns. That follow-up question and its results are presented in the image below.



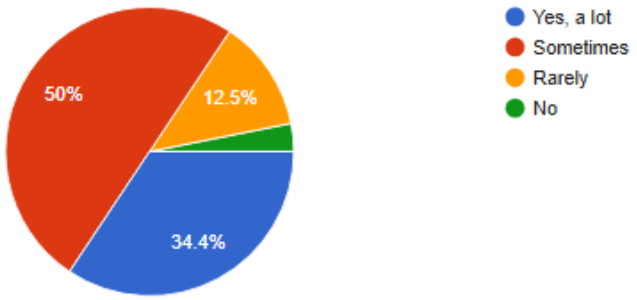




Do you usually have to wait in line to order food?

Copy chart

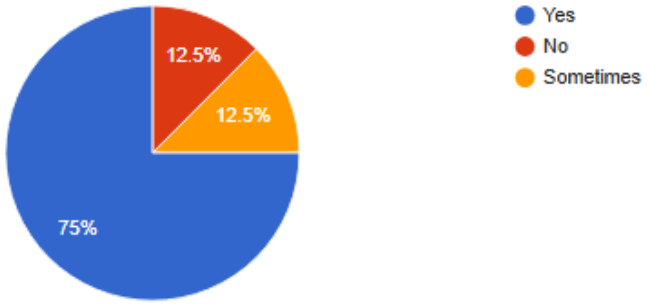
32 responses



Do you sometimes feel rushed when ordering because staff are waiting?

Copy chart

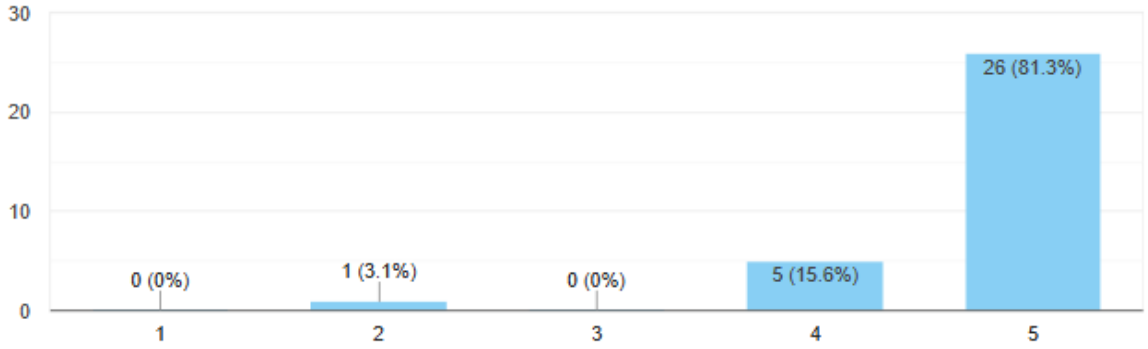
32 responses



I understand how QR codes work.

Copy chart

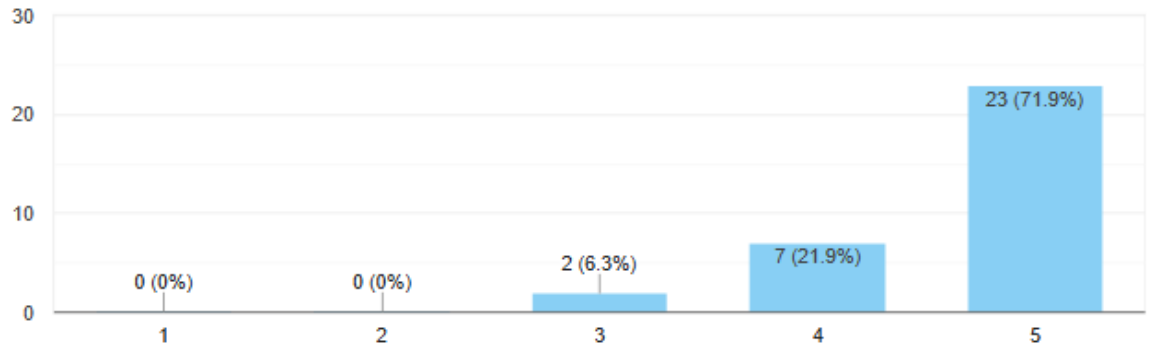
32 responses



I am comfortable using my phone to order food.

Copy chart

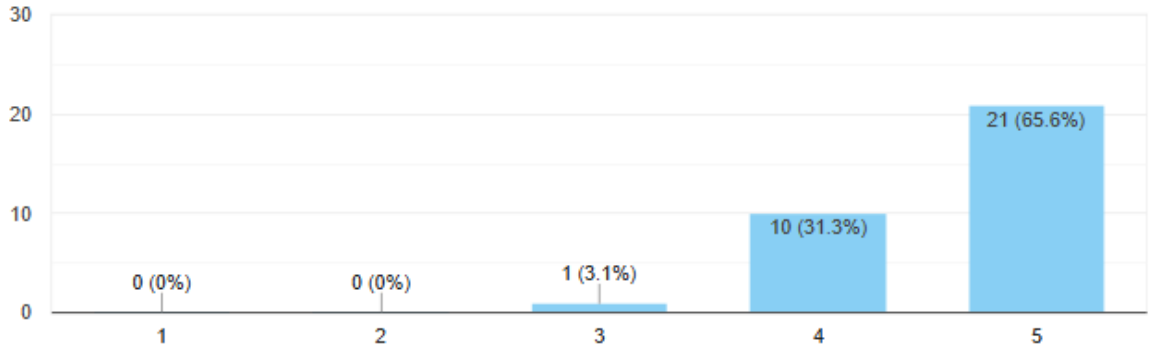
32 responses



I think using QR codes for menus could be useful.

Copy chart

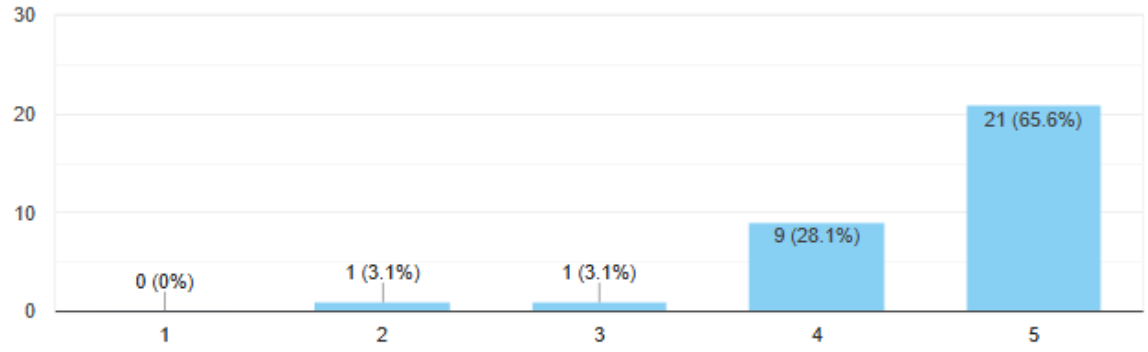
32 responses



I like the idea of ordering food from my seat.

Copy chart

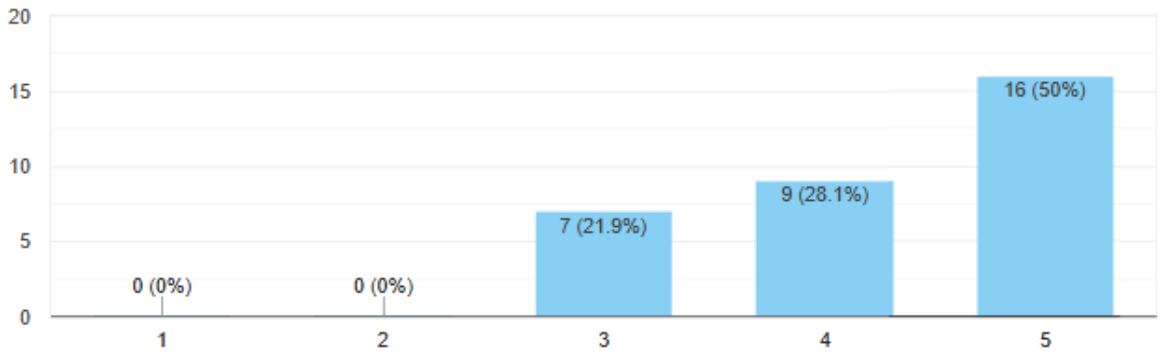
32 responses



Having food delivered to my table would be convenient.

Copy chart

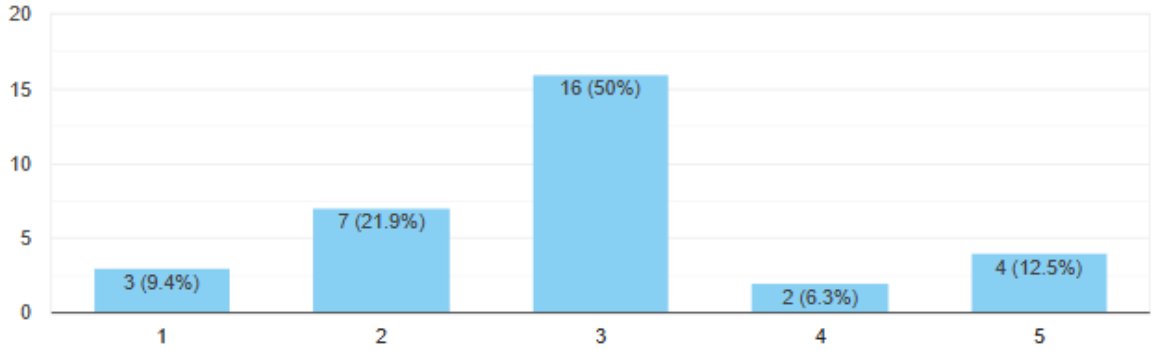
32 responses



I prefer to talk to staff when ordering instead of using my phone.

Copy chart

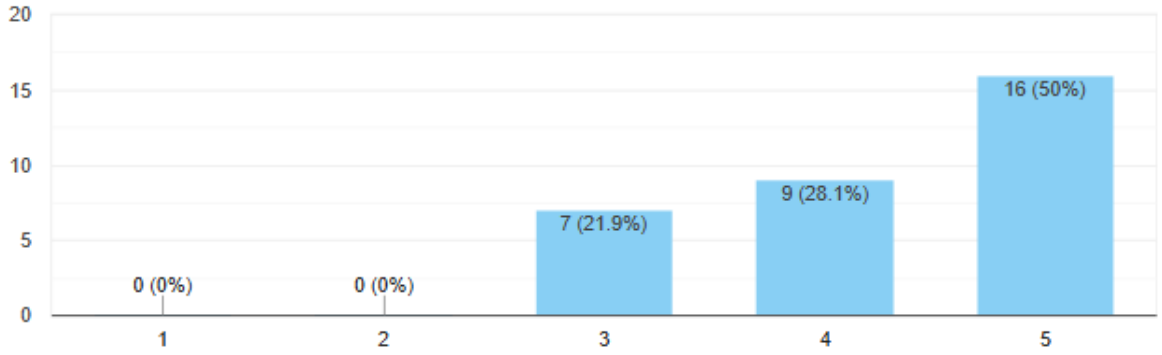
32 responses



Having food delivered to my table would be convenient.

Copy chart

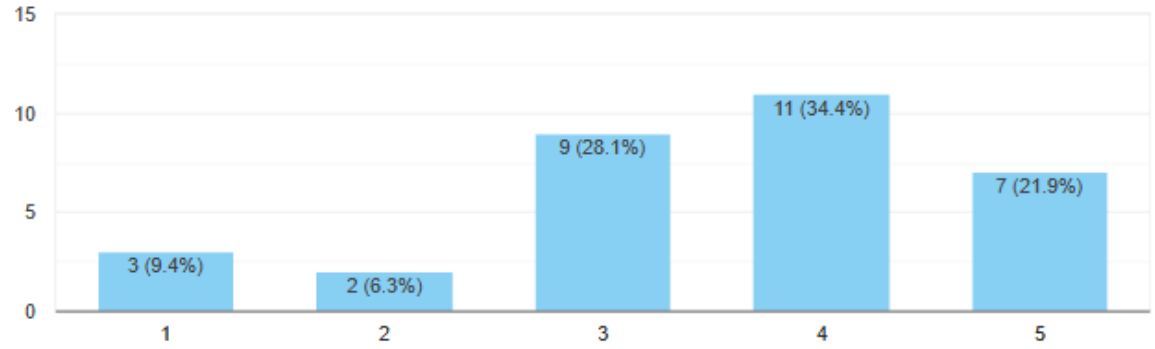
32 responses



I worry that I might face technical issues with the system.

Copy chart

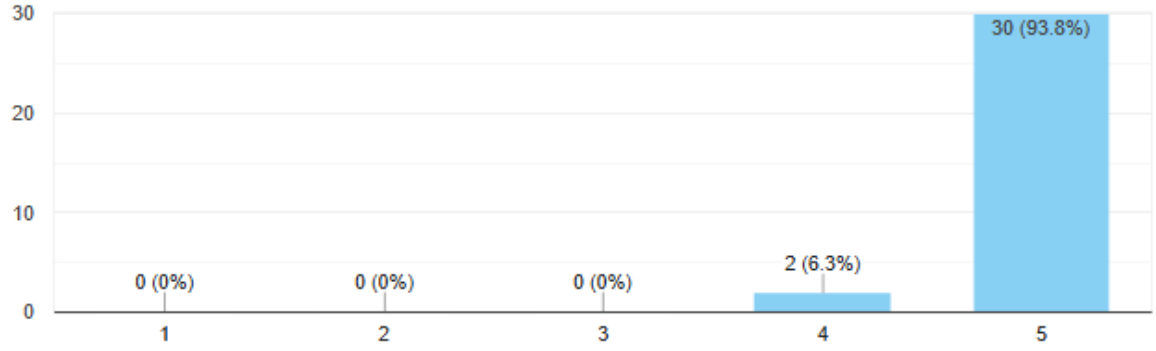
32 responses



I would like to see pictures, descriptions, and prices on the menu.

Copy chart

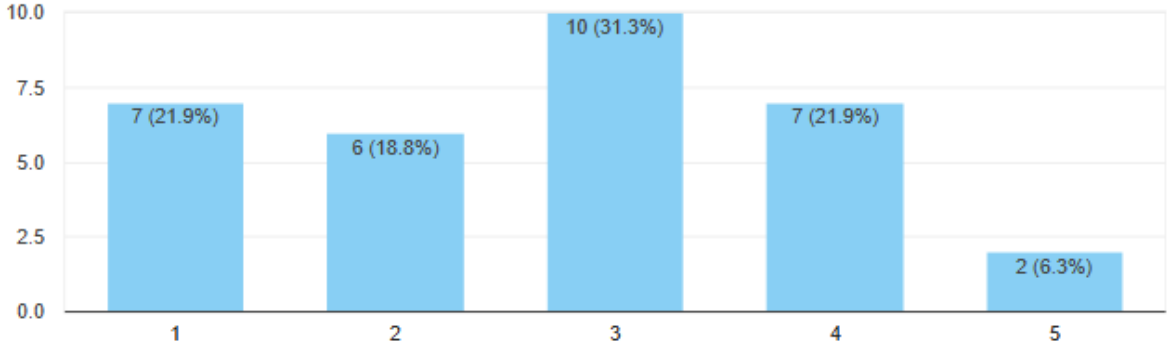
32 responses



I feel this system might make the ordering process more confusing.

Copy chart

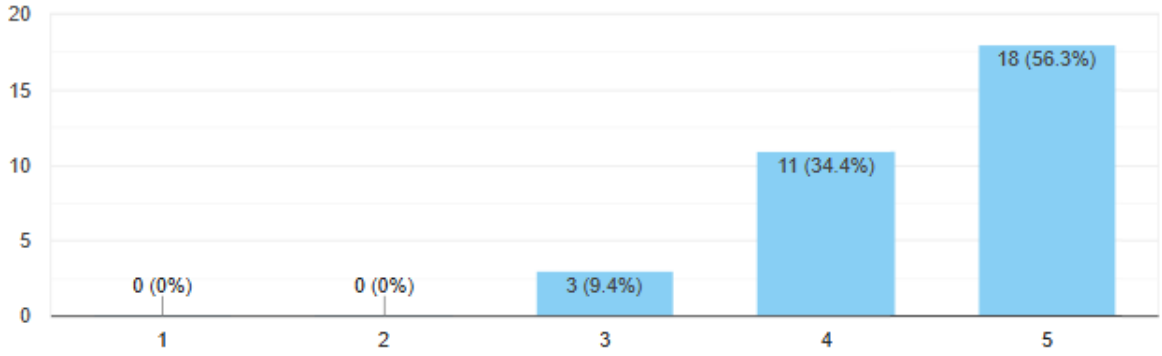
32 responses



A phone-based ordering system might make things faster.

Copy chart

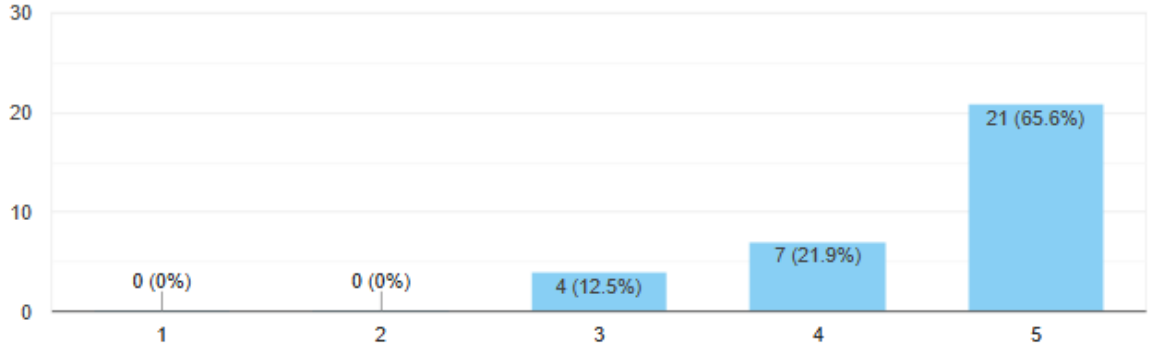
32 responses



I want the system to support both Khmer and English.

Copy chart

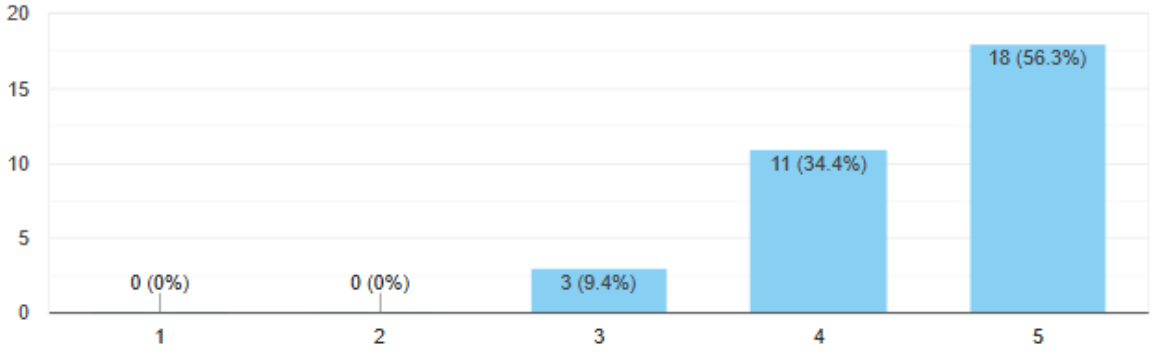
32 responses



A phone-based ordering system might make things faster.

Copy chart

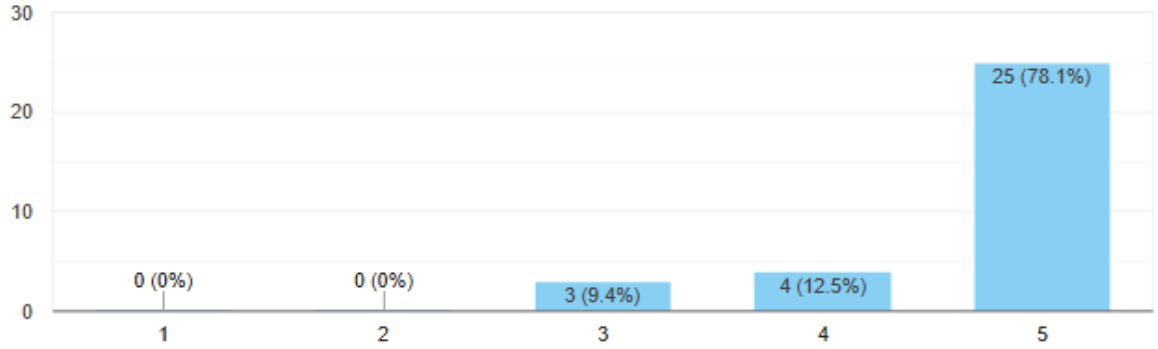
32 responses



I think this system is good for shy or quiet students.

Copy chart

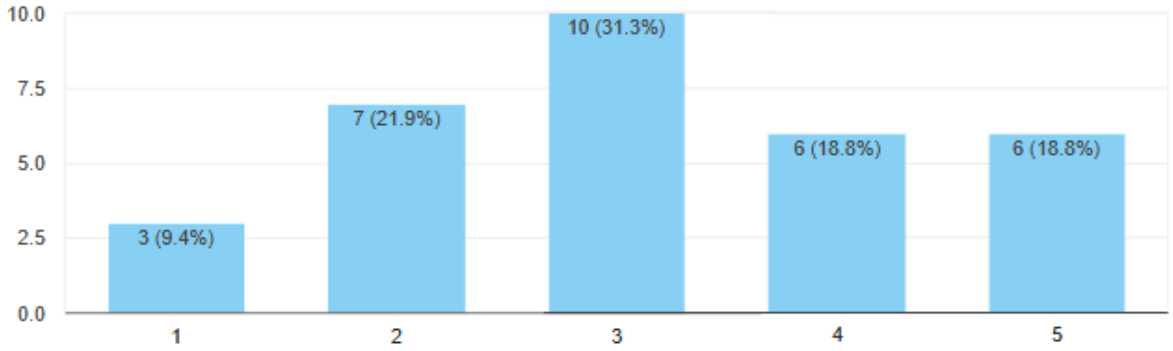
32 responses



I would need some help to use this kind of system.

Copy chart

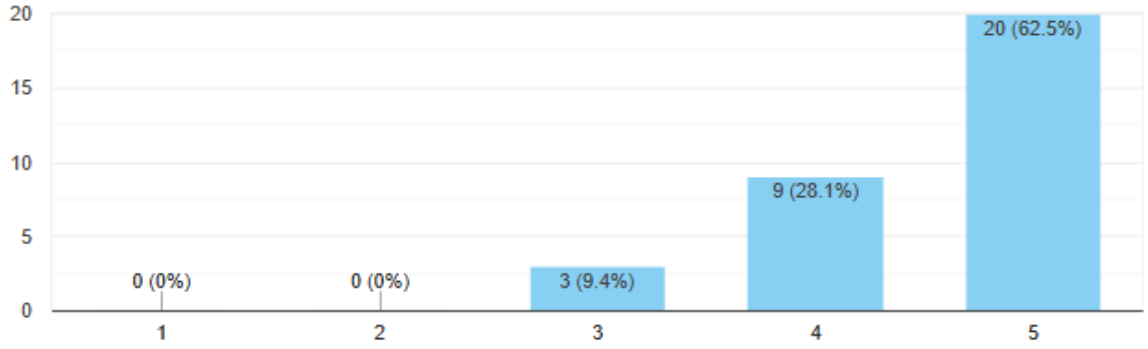
32 responses



I would like to try this system if it is available in our cafeteria.

Copy chart

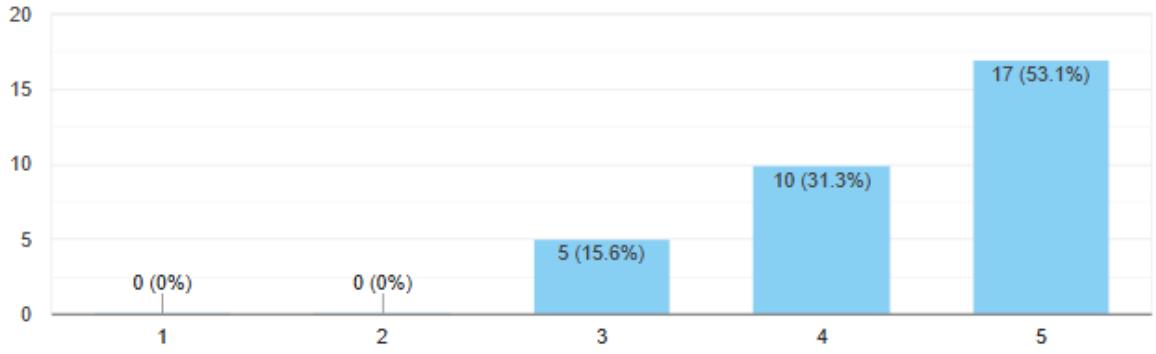
32 responses



I think it will make the cafeteria faster and more organized.

Copy chart

32 responses



I feel confident that I can use this system without needing help.

Copy chart

32 responses

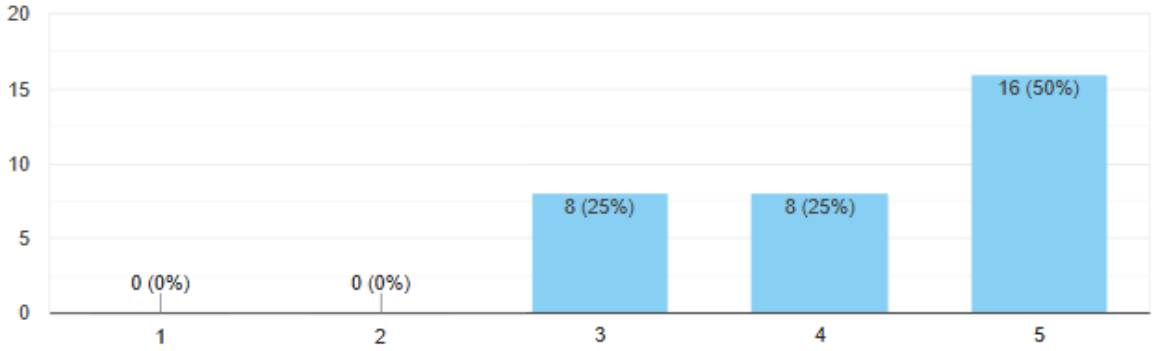


Figure 30: Result of all the Survey

6.4.6 Survey of the system from vendor

To assess the practical usability and impact of the QR Menu System, we conducted a structured interview with one of the participating food vendors. Their responses provided insight into the system's strengths, limitations, and areas for improvement based on real-world usage. Below is a detailed summary of their feedback:

1. Ease of Learning and Adoption

The vendor reported that the QR Menu System was fairly easy to learn. The interface was described as simple and intuitive, which allowed staff to begin using the system without extensive training. However, some staff members, particularly those less familiar with digital tools, required a short adjustment period to transition from paper-based order taking to a fully digital workflow. This indicates that while the system is user-friendly, initial onboarding and training may be necessary to ensure a smooth transition for all staff members.

2. Accuracy and Efficiency Improvements

Since adopting the system, the vendor observed noticeable improvements in order accuracy. This was attributed to the system's ability to display customer selections clearly, reducing the risk of human error that commonly occurs when taking orders verbally or writing them down manually. Additionally, efficiency in order handling improved, although it was noted that performance can be affected by slow internet connectivity. This suggests that while the digital system streamlines operations, it remains dependent on the reliability of internet infrastructure.

3. Perceived Benefits

The main benefits experienced by the vendor include:

- **Reduced crowding** at the food stall, as customers can order remotely via their smartphones.
- **Customer autonomy**, allowing users to place orders at their own pace without waiting in long lines.
- **Time savings** for the vendor, who no longer needs to repeatedly take verbal orders or manually record them.

These advantages contribute to a smoother customer experience and more efficient vendor operations, especially during peak hours.

4. Challenges and Issues Encountered

The primary challenge identified by the vendor is **internet reliability**. When the Wi-Fi connection is unstable or disconnected, the system may freeze or experience significant delays in loading and processing orders. This presents a major risk to service continuity. Furthermore, some older or less tech-savvy staff found it difficult to adapt to the digital system, indicating a need for targeted training or alternative support methods for such users.

5. Suggestions for Improvement

The vendor suggested two key enhancements to the current system:

- **Real-time notifications with audio alerts** when a new order is received. This would help staff respond more quickly without constantly monitoring the screen.
- **Offline functionality or backup method** to allow continued order taking when internet service is disrupted. This would ensure business continuity and reduce dependence on network stability.

These suggestions highlight areas where the system can evolve to better support vendors under different conditions.

6. Recommendation to Other Vendors

When asked whether they would recommend the QR Menu System to other vendors, the respondent answered positively—but with some conditions. They emphasized that the system works well **as long as internet connectivity is stable** and the menu does not change frequently. The vendor acknowledged that it is a **practical and effective solution overall**, but noted that vendors should be prepared for an initial learning curve, particularly in environments with limited digital literacy.

CHAPTER 7 : DISSCUSSION

The QR Code-Based Digital Menu System developed for the Royal University of Phnom Penh (RUPP) was designed to modernize the traditional food ordering experience within the university cafeteria by introducing a digital, contactless, and efficient solution. This system effectively addresses key pain points observed in conventional methods—such as slow service, communication errors, and long waiting times—by enabling users to scan a QR code and interact with a real-time digital menu on their personal devices.

One of the most impactful features of the system is its real-time menu management capability. Cafeteria vendors can instantly update food names, availability, and prices through the admin dashboard. These changes are reflected dynamically on the QR code-linked digital menus accessed by users, ensuring accuracy and reducing reliance on outdated printed menus. This enhances operational flexibility, especially during rush hours when frequent updates may be required.

The user interface (UI) and user experience (UX) design were developed using modern frontend technologies like Nuxt.js, Vue.js, and Tailwind CSS. The system supports both Khmer and English, making it accessible for a wide range of users. Feedback from system testing showed that over 90% of users found the platform intuitive and helpful, indicating a high level of user satisfaction. The mobile-first design ensures that students and staff can interact with the menu regardless of their device or screen size.

The backend infrastructure, built with Express.js and MongoDB Atlas, was selected for its flexibility in managing semi-structured data such as orders and menu items. The system uses secure authentication with JWT and implements AES-256 encryption protocols to lay the groundwork for future online payment integration. The use of cloud-based services like Vercel and Render allows for scalable deployment, while the architecture is modular enough to support future expansions.

However, the system is not without its challenges:

- Internet dependency remains a primary concern. Without stable campus-wide internet, users may experience loading delays or failures.

- Onboarding challenges were also observed—particularly among cafeteria staff who may not be tech-savvy. A future version may need to include training resources or a simplified backend interface for non-technical users.
- Reliance on manual payment is currently a limitation, as the platform does not yet support digital payments. This feature is planned for future implementation.

Two important features that have been planned for future development are:

1. Student Employment Module: A future version of the system will allow students to register for part-time delivery roles directly through the platform. This will provide flexible income opportunities and help students gain valuable soft skills such as time management, responsibility, and customer service—all within their university environment.
2. Supplier Integration System: The system is also designed to allow local departments (e.g., Agriculture) to apply as official food suppliers. In the future, these departments could provide fresh produce to the cafeteria, creating a sustainable food supply chain and giving students hands-on experience in business operations, logistics, and product marketing.

These planned features are expected to extend the system's impact beyond digital ordering into broader community engagement, employment, and academic collaboration.

In conclusion, while the current version of the system successfully streamlines food ordering and improves operational efficiency within RUPP, its full potential will be realized in future versions—when features like digital payments, student job integration, and supplier management are implemented. The system lays a solid foundation for a comprehensive, campus-wide smart cafeteria platform that aligns with modern digital practices and supports both operational needs and educational goals.

CHAPTER 8 : CONCLUSION

The QR Code-Based Digital Menu System demonstrates how digital transformation can be successfully applied in educational environments to improve service delivery, reduce paper waste, and foster student involvement. By leveraging QR technology, modern web tools (Nuxt.js, MongoDB, Express.js), and feedback-driven design, the system has enhanced the food ordering experience for both customers and vendors.

It offers a scalable, eco-friendly, and user-centric platform that not only streamlines cafeteria operations but also integrates student employment and localized supplier collaboration. While technical and operational challenges remain, the positive user feedback and functional system prototype indicate strong potential for future expansion.

In conclusion, this project not only fulfills its core objectives—such as reducing wait times and enhancing operational efficiency—but also lays the groundwork for further innovation in campus-based digital services. With continued support and development, it has the capacity to become a model for digital menu systems in similar academic settings nationwide.

CHAPTER 9 : FUTURE PLAN

The future development of the QR Code Menu project aims to elevate both user and vendor experiences through targeted enhancements, prioritizing scalability, usability, and operational efficiency. Key initiatives include:

1. **Secure Payment Integration**: A dedicated payment page will be developed to enable seamless and secure transactions. This will support popular payment methods in Cambodia, such as mobile payment apps (e.g., Wing, Pi Pay , Bakong) and QR code-based payments, ensuring compliance with industry-standard encryption protocols (e.g., PCI DSS). This feature will enhance user convenience and trust while streamlining revenue collection for vendors.
2. **Enhanced Delivery Management and Job Opportunities**: A robust delivery assignment system will be introduced, allowing vendors to efficiently recruit and manage student delivery personnel via an integrated dashboard. Vendors can post delivery tasks, assign them based on availability and proximity, and track performance metrics (e.g., delivery times). A job opportunities for students will be advertised through the RUPP QR-Menu platform and university channels, targeting at least 20 part-time delivery roles initially. This will optimize logistics, reduce delivery delays, and provide students with flexible, skill-building job opportunities that foster time management, communication, and customer service skills.
3. **Real-Time Order Tracking**: The system will incorporate WebSocket technology to provide real-time order status updates, notifying users when their order is received, in preparation, ready for pickup, or out for delivery. A user-friendly interface will display these updates, reducing uncertainty and enhancing customer satisfaction by keeping users informed throughout the ordering process.
4. **Performance Optimization and Scalability**: To accommodate growing user demand, the system will undergo performance optimizations, including load balancing, database indexing enhancements, and caching mechanisms (e.g., Redis for frequently accessed menu data). These improvements will ensure responsiveness during peak hours and enable scalability for potential expansion to other university campuses or institutions.

5. **User Feedback and Analytics**: A feedback module will be implemented, allowing users to rate their experience and provide comments on food quality, delivery, and service. Vendors will access aggregated analytics through the dashboard to identify trends, improve offerings, and address pain points. This data-driven approach will foster continuous improvement and user-centric development.
6. **Supplier Opportunities for Campus Departments and Local Vendors**: The system will expand its supplier integration feature, enabling university departments (e.g., Faculty of Agricultural Sciences) and local vendors to register as official suppliers of fresh produce or other goods.A supplier application portal will be launched, allowing departments and vendors to submit product lists and pricing for review by the cafeteria team. Approved suppliers will manage their listings via a dedicated dashboard, promoting sustainability and supporting campus-driven food sourcing. This initiative will provide hands-on experience for agriculture and business students in supply chain management and marketing, with an initial goal of onboarding at least five campus-based or local suppliers.

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Source code for Frontend and Backend

Frontend

https://github.com/cheakimhengg/Nhameys

https://github.com/cheakimhengg/Celastic

Backend

https://github.com/phannyvath/Qr-Menu-Backend